GCEMNet: Grouped Cascaded Enhanced Mamba Network for Hyperspectral Image Classification in Spectrally Similar Scenarios

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***Abstract*—Hyperspectral image (HSI) classification remains challenging in spectrally similar scenarios, where different ground objects may share highly overlapping spectral signatures and comparable spatial structures. In such cases, weak discriminative cues are easily obscured by redundant spectral and spatial responses, leading to confusion between categories with similar material composition or spatial patterns. To address this problem, this article proposes a grouped cascaded enhanced Mamba network (GCEMNet) for HSI classification in spectrally similar scenarios. GCEMNet follows a progressive feature learning pipeline. Specifically, the channel–spatial attention fusion (CSAF) module preserves subtle discriminative cues by jointly exploiting channel statistics and spatial structural information; the grouped cascaded Mamba fusion (GCMF) module, built upon multiple enhanced Mamba (EM) units, captures complementary local and global dependencies through grouped learning and cascaded propagation; and the cross-feature separation Transformer (CFST) reduces the influence of redundant spectral and spatial responses through correlation-aware feature modeling and cross-feature interaction. Extensive experiments on two public mineral HSI datasets and one agricultural HSI dataset demonstrate that GCEMNet achieves superior classification accuracy with moderate model complexity. Module comparison, ablation studies, visualization results, and computational complexity analysis further verify the effectiveness and practicality of the proposed modules in spectrally similar scenarios and scenarios with class imbalance.**

***Index Terms*—Cross-feature separation Transformer (CFST), enhanced Mamba (EM), feature fusion, hyperspectral image (HSI) classification, spectral similarity.**

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I. Introduction

HYPERSPECTRAL imaging has become an important technology in Earth observation and remote sensing, as it captures both the spatial morphology of ground objects and their continuous spectral signatures across hundreds of narrow, contiguous bands [1]. As a fundamental task in hyperspectral image (HSI) analysis, HSI classification establishes a critical link between raw hyperspectral data and actionable geospatial decision-making by assigning pixels to predefined land-cover categories. It has been widely applied in environmental monitoring, mineral exploration, precision agriculture, vegetation health assessment, lithological mapping, and crop stress diagnosis [2], [3], [4].

However, hyperspectral data are inherently characterized by high dimensionality and information ambiguity. Different ground objects may exhibit highly similar spectral signatures, whereas the same object may show significant spectral variability under different environmental conditions. These factors impose challenges on both computational efficiency and feature extraction accuracy. To address these issues, various dimensionality reduction and information enhancement methods have been developed. Representative dimensionality reduction techniques include principal component analysis (PCA) [5], independent component analysis (ICA) [6], linear discriminant analysis (LDA) [7], and minimum noise fraction (MNF) transformation [8]. Information enhancement methods include extended morphological attribute profiles (EMAPs) [9], morphological profiles (MPs)/enhanced MPs (EMPs) [10], SLIC superpixel segmentation [11], and bilateral filtering (BF) [12]. These methods improve spectral–spatial feature discriminability and provide useful low-dimensional representations for subsequent classification models.

Early HSI classification methods mainly relied on handcrafted features and classical classifiers, such as support vector machines (SVMs) [13], random forests (RFs) [14], and K-nearest neighbors (KNNs) [15]. Among them, SVM-based methods achieved strong robustness in high-dimensional feature spaces, and subsequent studies further improved classification performance through weighted kernels, adaptive

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feature selection, and feature fusion strategies [16], [17], [18], [19]. Nevertheless, these traditional methods heavily depend on manually designed features and prior expertise, limiting their generalization ability in complex land cover scenarios with spectral similarity and variability.

Deep learning has significantly advanced HSI classification by enabling automatic feature representation learning. CNN-based models have been widely explored. In general, 1-D CNNs focus on spectral feature extraction, 2-D CNNs emphasize spatial structure modeling, and 3-D or hybrid CNNs jointly use spectral and spatial information [20], [21], [22]. Despite their effectiveness, CNNs are still constrained by local receptive fields and may have difficulty modeling dependencies over long ranges and interactions between spectral and spatial features in complex scenes.

Fusion of spectral and spatial features has therefore become a central topic in HSI classification [23]. Existing fusion architectures can be broadly divided into serial, parallel, and cross-forms. Serial architectures may lose useful information during staged processing, parallel architectures often provide limited interaction between branches, and cross-architectures improve dynamic fusion but usually introduce higher computational cost and weaker noise robustness [20], [21], [24]. In addition, many fusion models depend on large numbers of labeled samples and may suffer from parameter redundancy or low inference efficiency, especially in small-sample and real-time classification scenarios [25], [26].

To overcome these limitations, advanced architectures such as Transformers, graph convolutional networks (GCNs), and state-space models (SSMs), represented by Mamba [27], have been introduced into HSI classification. Transformers are effective in modeling long-range dependencies but incur quadratic computational complexity. GCNs are suitable for structured non-Euclidean feature modeling but depend heavily on adjacency construction. Mamba naturally fits continuous spectral sequence modeling with linear computational complexity. Although recent Mamba-based HSI models have shown promising potential [28], [29], their ability to distinguish highly similar spectral–spatial categories remains insufficient, motivating the design of more targeted hybrid architectures.

Spectrally similar scenarios are pervasive in precision agriculture, mineral exploration, wetland conservation, and other remote sensing applications. These scenarios are characterized by highly overlapping spectral signatures, tightly coupled spectral and spatial responses, and subtle micro-textural variations among ground objects, such as spectral confusion between closely related crop species or spectral overlap between symbiotic minerals. Such characteristics remain challenging for existing HSI classification models. First, weak discriminative cues are easily weakened during shallow spectral–spatial fusion, especially when parallel branches only perform simple concatenation or weighted aggregation. Second, long-range spectral dependencies are difficult to model efficiently: CNNs are limited by local receptive fields, while Transformers usually suffer from high computational complexity for high-dimensional HSI data. Although Mamba provides an efficient alternative for long-sequence modeling, single-stream Mamba structures may still be insufficient to capture complementary channel-wise and spatial dependencies. Third, redundant spectral and spatial responses are not explicitly separated in many existing models, which may obscure subtle differences among spectrally similar categories. As summarized in Fig. 1, direct spectral–spatial fusion may mix redundant responses, while shallow fusion may provide insufficient interaction; this motivates feature-enhanced spectral–spatial fusion with selective sequential modeling.

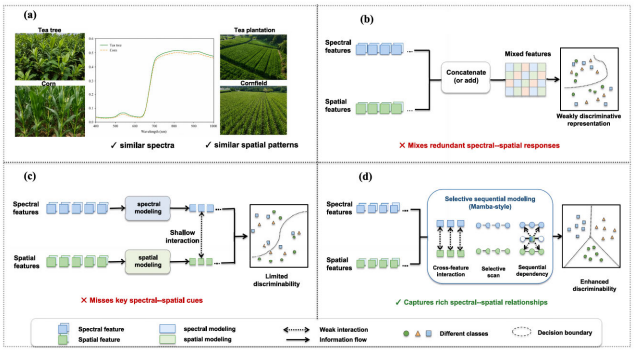


Fig. 1. Motivation of feature-enhanced spectral–spatial fusion in spectrally similar scenarios. (a) Similar crop categories often share close spectral signatures and spatial patterns. (b) Direct fusion may mix redundant spectral–spatial responses and produce weak representations. (c) Shallow fusion provides limited spectral–spatial interaction and may miss key cues. (d) Feature-enhanced fusion strengthens cross-feature interaction and selective sequential modeling, yielding more discriminative representations.

To address these challenges, this article proposes a grouped cascaded enhanced Mamba network (GCEMNet) for HSI classification in spectrally similar scenarios. GCEMNet follows a progressive feature learning pipeline consisting of spectral–spatial fusion extraction, grouped cascaded Mamba enhancement, and correlation-aware spectral–spatial refinement. Correspondingly, channel–spatial attention fusion (CSAF) preserves subtle discriminative cues during shallow feature extraction by jointly exploiting channel statistics and spatial structural information. Grouped cascaded Mamba fusion (GCMF), built upon multiple enhanced Mamba (EM) units, models complementary local and global dependencies through grouped feature learning, global contextual modeling, and cascaded propagation. The subsequent spectral-guided refinement further allows channel-derived spectral cues to guide spatial sequence modeling. Finally, the cross-feature separation Transformer (CFST) models spectral and spatial correlations separately and performs cross-feature interaction to reduce redundant responses and enhance complementary discriminative information.

In summary, the key contributions of this work are as follows.

1) We propose GCEMNet, a GCEMNet tailored for HSI classification in spectrally similar scenarios. It progressively integrates spectral–spatial fusion extraction, grouped cascaded Mamba enhancement, and matrix-guided feature separation to capture subtle discriminative cues among highly confused categories.

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2) We design a multistage spectral–spatial feature learning pipeline composed of CSAF, EM, and GCMF. CSAF enhances shallow channel–spatial interactions, EM models channel-wise and spatial sequential dependencies, and GCMF aggregates grouped local–global contextual information through cascaded propagation.
3) We introduce CFST for matrix-guided feature separation and cross-feature interaction. Spectral and spatial adjacency matrices are constructed to obtain correlation-aware separated features, which are further refined by bidirectional cross-feature interaction to reduce the influence of redundant responses and enhance complementary information.
4) Extensive experiments on two mineral datasets and one agricultural dataset demonstrate the effectiveness and robustness of GCEMNet in spectrally similar scenarios and scenarios with class imbalance. Additional module comparison, ablation, visualization, and computational complexity analyses further verify the practicality of the proposed method.

The remainder of this article is organized as follows. Section II reviews related work. Section III introduces the proposed GCEMNet. Section IV presents the experimental results and analysis. Finally, Section V concludes this article and discusses future research directions.

II. Related Work

This section reviews representative advances related to HSI classification from three aspects: spectral–spatial attention mechanisms, Mamba-based feature enhancement, and graph-based adjacency modeling. For each direction, we summarize its design motivation and discuss its limitations in distinguishing spectrally similar ground objects, thereby clarifying the motivation of the proposed GCEMNet.

A. Spectral–Spatial Attention

Spectral and spatial attention mechanisms dynamically assign feature weights to enhance informative responses and suppress irrelevant noise and have become important techniques for spectral–spatial fusion in HSI classification. They compensate for the limited local receptive fields of conventional convolutional networks and improve feature discriminability, evolving from single-dimensional modeling, such as spectral-only or spatial-only attention, to joint spectral–spatial attention paradigms.

The squeeze-and-excitation (SE) module [30] pioneered adaptive channel-wise weighting via global pooling, capturing interchannel dependencies and laying the foundation for subsequent attention research. CBAM extended this by adding a spatial branch, achieving dual enhancement of channel and spatial features. Sun et al. [31] proposed a dual-branch network focusing on key spectral bands and target regions to suppress heterogeneous pixel interference. Hang et al. [32] improved classification stability in small-sample scenarios through dedicated branch design.

Zhu et al. [33] embedded spectral and spatial attention into residual structures to enhance feature representation, but their method was not specifically designed for spectrally similar ground objects. He et al. [28] developed a dual-attention module to capture global and local features; however, the fusion between complementary spectral and spatial responses remains limited in complex spectral scenarios. Shi et al. [34] proposed a multiresolution fusion network, while its capability in extracting subtle microtextural differences among similar objects still needs further improvement.

Wang et al. [35] introduced nonlocal operations as a versatile building block for capturing long-range dependencies in computer vision, achieving state-of-the-art performance in video classification and improving tasks such as object detection and pose estimation. Zhang et al. [36] later proposed the residual nonlocal attention network (RNLAN) for image restoration, addressing limitations of methods that treat spatial and channel-wise features equally and rely solely on local convolutions.

Although existing spectral–spatial attention methods have improved feature representation, most of them still rely on relatively independent channel and spatial modeling or shallow feature fusion. As a result, the interaction between complementary spectral and spatial responses remains insufficient, especially when different categories exhibit highly similar spectral signatures. To address this limitation, the proposed CSAF module enhances shallow spectral–spatial representations through collaborative channel–spatial attention learning, providing a more reliable foundation for subsequent long-range dependency modeling and fine-grained feature enhancement.

B. Mamba-Based Feature Enhancement

Rooted in SSMs, Mamba has attracted increasing attention as an efficient sequence modeling architecture. Compared with Transformer self-attention, Mamba offers linear computational complexity and strong long-sequence modeling capability, which naturally aligns with the continuous spectral structure of HSIs. Therefore, Mamba-based models have become a promising direction for HSI processing, as they can efficiently capture long-range dependencies through selective state-space modeling and input-dependent parameterization.

For HSI classification, He et al. [28] proposed a lightweight spectral and spatial Mamba architecture by combining Mamba with convolutional modules for efficient spectral sequence extraction, but its generalization ability remains limited in complex spectrally similar scenarios. Huang et al. [29] designed a spectral–spectral Mamba network to enhance spectral sequence correlations, but the deep coupling between spatial and spectral features was not fully explored. Ma et al. [37] fused CNN local perception and Mamba global modeling in a dual-branch structure, while dedicated enhancement for microtextural differences among similar objects remains insufficient.

Compared with these representative Mamba-based HSI methods, GCEMNet introduces several targeted designs for spectrally similar scenarios. EM performs bidirectional Mamba modeling along both channel and spatial sequences with adaptive feature fusion. Based on EM, GCMF further combines group-wise feature modeling, a parallel global

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contextual branch, and progressive cascaded propagation to strengthen local–global interaction. After GCMF, the spectral-guided refinement establishes cross-guidance between channel and spatial representations, allowing channel-derived spectral cues to guide spatial sequence modeling.

C. Graph Convolution and Adjacency Matrix Optimization

GCNs excel at modeling non-Euclidean spatial features and characterizing node correlations via adjacency matrices, enabling structured modeling of HSI spectral and spatial features. The construction and optimization of adjacency matrices directly determine the performance of feature aggregation and redundancy elimination.

Screening-based feature separation usually relies on static rules to select features, making it difficult to model dynamic spectral and spatial correlations and preserve subtle discriminative cues. Projection-based dimensionality reduction compresses feature dimensions, but may weaken spectral–spatial topological structures and reduce sensitivity to fine-grained feature differences. In contrast, GCNs establish node correlations through adjacency matrices, enabling structured feature aggregation while preserving inherent spectral and spatial relationships. Therefore, adjacency matrix construction plays a critical role in graph-based HSI feature modeling.

Wan et al. [38] constructed superpixel-level graphs using the SLIC algorithm, using superpixel feature means as node features and fixed adjacency matrices to reduce computational cost, but failed to capture dynamic feature correlations. Hong et al. [39] proposed mini-GCN, which optimizes adjacency matrix efficiency via batch-wise subgraph division, but causes spectral and spatial information confusion by ignoring dimension independence. Chen et al. [40] combined spectral-guided superpixel segmentation with graph Transformers, but lacked dedicated redundant information elimination for fused features. Graph attention methods also suffer from matrix distortion in spectrally similar scenarios due to feature redundancy.

Existing graph-based methods still face two major limitations. First, fixed or weakly adaptive adjacency matrices are insufficient to distinguish effective spectral–spatial correlations from redundant responses in spectrally similar scenarios. Second, most methods lack an explicit separation-and-interaction mechanism, which limits the utilization of dimension-specific spectral and spatial information. To address these problems, the proposed matrix feature separator (MFS) adaptively constructs spectral and spatial adjacency matrices to obtain dimension-specific correlation-aware representations, while the cross-feature Transformer (CFT) further performs bidirectional interaction between the refined features for complementary information enhancement.

III. Proposed Method

Fig. 2(a) illustrates the overall architecture of GCEMNet. The network consists of three progressive stages: spectral–spatial fusion extraction, grouped cascaded Mamba enhancement, and matrix-guided feature separation. Given a PCA-reduced local HSI patch, the first stage extracts shallow spectral and spatial representations using convolutional projection and CSAF. The second stage employs GCMF, which is built upon multiple EM units, to perform group-wise spectral–spatial enhancement and cascaded local–global feature aggregation. The third stage uses CFST to construct correlation-aware separated representations and perform cross-feature interaction for redundancy suppression and complementary information enhancement.

In this work, PCA and local patch extraction are used as standard input construction steps rather than learnable components of the proposed model. Specifically, PCA is first applied to the original HSI to reduce spectral redundancy. Then, for each labeled pixel, a local patch centered at that pixel is extracted to incorporate spatial neighborhood information. The resulting input tensor is denoted as $\mathbf{X} \in \mathbb{R}^{N\times C\times P\times P}$, where N, C, and P denote the batch size, the number of retained PCA components, and the patch size, respectively. Based on this input representation, GCEMNet learns discriminative spectral–spatial features through the following network modules.

A. Spectral–Spatial Fusion Extractor

1) Channel–Spatial Attention Fusion: Existing spectral–spatial attention mechanisms often rely on independent channel and spatial modeling, resulting in limited interaction between complementary feature responses. To alleviate this issue, the CSAF module is introduced to enhance shallow spectral–spatial representations through collaborative channel and spatial attention learning.

As shown in Fig. 3, the CSAF module adopts a dual-branch parallel attention mechanism, including a channel attention branch and a spatial attention branch. Let the input feature be $\mathbf{F} \in \mathbb{R}^{N\times C_f\times P\times P}$, where N is the batch size, C_f denotes the number of feature channels, and $P \times P$ represents the spatial size of the input patch. For notational simplicity, $\mathbf{F}$ is denoted as $\mathbf{X}$ in the following equations.

For channel attention, complementary global and local descriptors are constructed using mean, max, and standard-deviation statistics, where the standard-deviation descriptor captures spectral fluctuation information often overlooked by conventional attention mechanisms

$$\begin{aligned}\mathbf{X}_c^g &= \mathbf{X}\odot \mathrm{Conv}_{1\text{-D}}^{g}\left(\mathrm{cat}\left(\mathrm{Mean}_{sp}\left(\mathbf{X}\right),\mathrm{Std}_{sp}\left(\mathbf{X}\right)\right)\right)\\ \mathbf{X}_c^l &= \mathbf{X}\odot \mathrm{Conv}_{1\text{-D}}^{l}\left(\mathrm{cat}\left(\mathrm{Max}_{sp}\left(\mathbf{X}\right),\mathrm{Std}_{sp}\left(\mathbf{X}\right)\right)\right).\end{aligned} \tag{1}$$

To exploit the complementarity between global and local descriptors, a bidirectional cross-fusion strategy is adopted

$$\begin{aligned}\mathbf{Z}_c^g &= \mathrm{Softmax}_c\left(\mathrm{Mean}_{sp}\left(\mathbf{X}_c^g\right)\right)\odot \mathbf{X}_c^l\\ \mathbf{Z}_c^l &= \mathrm{Softmax}_c\left(\mathrm{Mean}_{sp}\left(\mathbf{X}_c^l\right)\right)\odot \mathbf{X}_c^g\\ \mathbf{A}_c &= \sigma\left(\mathrm{Adp}_1\left(\mathrm{Conv}_{2\text{-D}}^{c}\left(\mathcal{R}\left(\mathrm{cat}\left(\mathbf{Z}_c^g,\mathbf{Z}_c^l\right)\right)\right)\right)\right).\end{aligned} \tag{2}$$

For spatial attention, a multiscale spatial descriptor is constructed as

$$\mathbf{M}_s = \mathcal{U}\left(\mathrm{Adp}_2\left(\mathbf{X}\right)\right)\odot \mathcal{U}\left(\mathrm{Adp}_3\left(\mathbf{X}\right)\right). \tag{3}$$

Based on the multiscale descriptor, two complementary spatial interaction paths are constructed

$$\mathbf{Z}_s^1 = \mathrm{Conv}_{2\text{-D}}^{s,1}\left(\mathbf{X}\right)\odot \mathrm{Softmax}\left(\mathbf{X}\odot \mathrm{Adp}_1\left(\mathbf{M}_s\right)\right)$$

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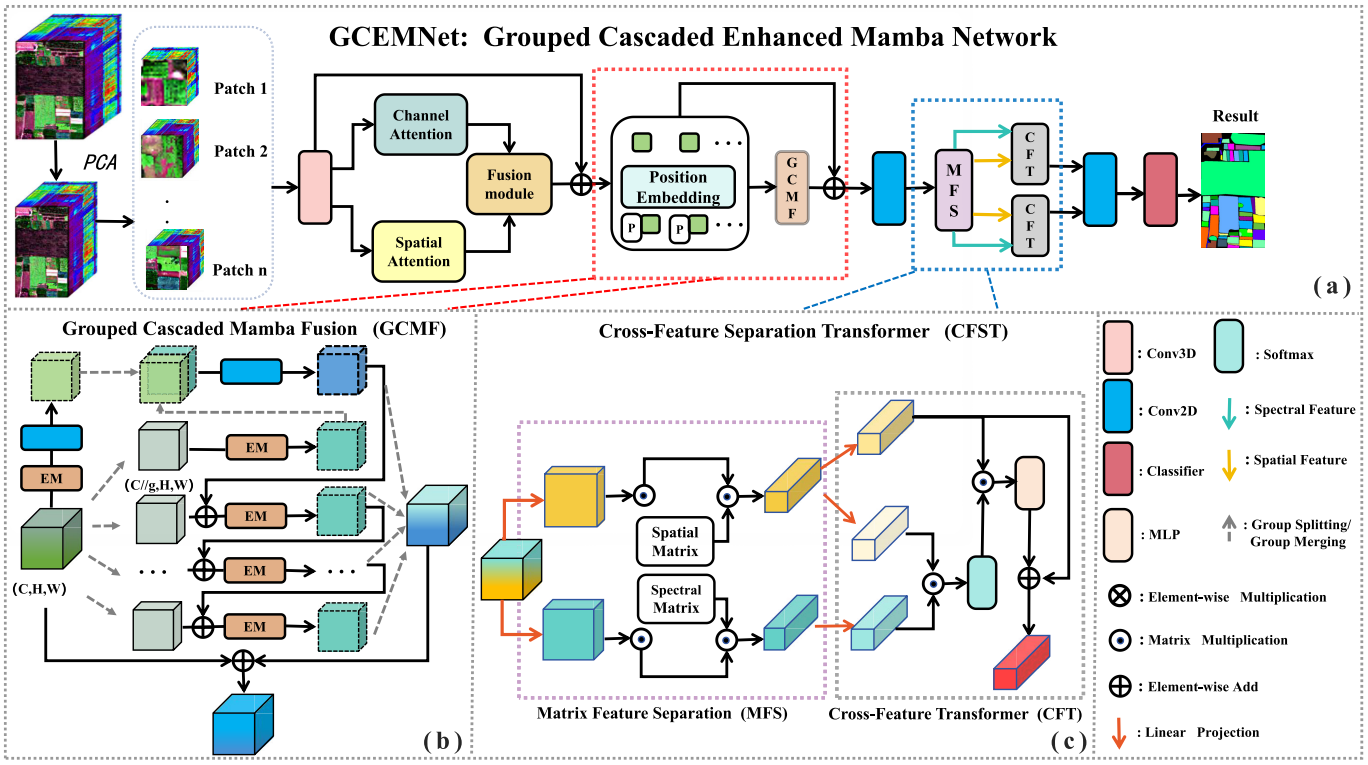


Fig. 2. Proposed GCEMNet architecture. (a) Overall network structure for HSI classification, where CSAF, GCMF, and CFST are progressively integrated to enhance the discrimination of spectrally similar categories. (b) GCMF module, which performs intergroup cascading and intragroup fusion through multiple EM units to generate more discriminative spectral and spatial representations. (c) CFST module, where the MFS constructs spectral and spatial adjacency matrices for redundancy suppression, and the CFT performs cross-feature interaction to obtain correlation-enhanced features.

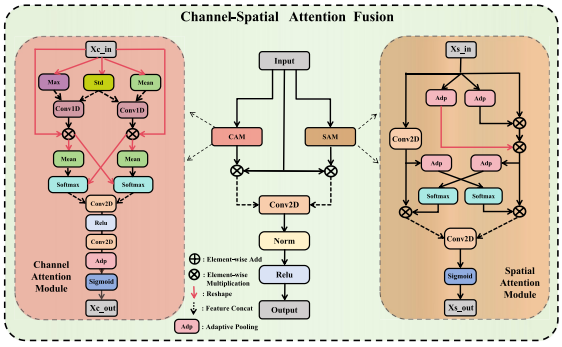


Fig. 3. CSAF module. It uses dual-branch parallel attention and residual fusion to capture spectral–spatial correlations, extract subtle features, and reduce information loss in shallow fusion.

$$\mathbf{Z}_s^2 = \mathbf{X} \odot \mathbf{M}_s \odot \mathrm{Softmax}\left(\mathrm{Adp}_1\left(\mathrm{Conv}_{2\text{-D}}^{s,2}(\mathbf{X})\right)\right)$$
$$\mathbf{A}_s = \sigma\left(\mathrm{Conv}_{2\text{-D}}^{s,0}\left(\mathrm{cat}\left(\mathbf{Z}_s^1, \mathbf{Z}_s^2\right)\right)\right). \quad (4)$$

Finally, the channel and spatial enhanced features are fused to generate the CSAF output

$$\mathbf{X}_{\mathrm{csaf}} = \mathcal{R}\left(\mathrm{Norm}\left(\mathrm{Conv}_{2\text{-D}}^{o}\left(\mathrm{cat}\left(\mathbf{X} \odot \mathbf{A}_s, \mathbf{X} \odot \mathbf{A}_c\right)\right)\right)\right). \quad (5)$$

Unlike conventional attention modules that treat channel and spatial responses separately, CSAF builds interactions between global and local channel statistics and multiscale spatial structures. The standard deviation descriptor preserves spectral fluctuation cues, while the spatial branch retains local structural consistency. This design gives the following Mamba modules a cleaner and more discriminative shallow representation.

Here, σ, $\mathcal{R}$, $\odot$, cat, and Norm denote the Sigmoid function, ReLU activation, element-wise multiplication, feature concatenation, and layer normalization, respectively. Mean, Std, and Max denote statistical pooling operations, while $\mathrm{Conv}_{1\text{-D}}$, $\mathrm{Conv}_{2\text{-D}}$, Adp_k, and $\mathcal{U}$ denote learnable convolutions, adaptive pooling, and upsampling operations, respectively.

2) Enhanced Mamba: While CSAF effectively preserves discriminative shallow spectral–spatial cues, the resulting features are still dominated by local receptive-field interactions and lack explicit modeling of long-range sequential dependencies. This limitation becomes particularly pronounced when distinguishing ground objects with highly overlapping spectral signatures. To further enhance fine-grained spectral–spatial discrimination, an EM unit is introduced. EM performs bidirectional Mamba modeling along both channel and spatial sequences and adaptively fuses the resulting representations, thereby jointly capturing channel-wise spectral dependencies and spatial contextual relationships. EM serves as the basic Mamba fusion unit for both shallow spectral–spatial sequence modeling and subsequent grouped cascaded enhancement, as illustrated in Fig. 4.

Given the CSAF-enhanced feature $\boldsymbol{X}_{\mathrm{m_in}} \in \mathbb{R}^{N \times C_f \times P \times P}$, the proposed EM unit performs dual-path sequence modeling from both channel and spatial perspectives. By transforming spectral and spatial features into sequential representations, the proposed unit enables Mamba to capture long-range

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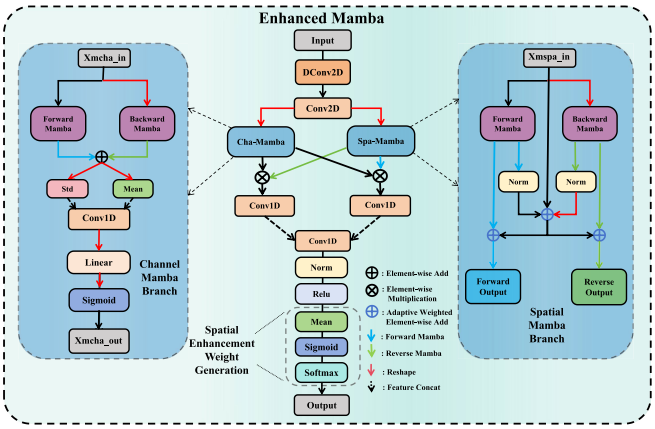


Fig. 4. EM module. It contains channel and spatial Mamba branches with bidirectional sequence modeling and adaptive feature fusion for feature enhancement.

dependencies that are difficult to model using conventional convolutional operations. The reshaped inputs of the channel and spatial Mamba branches are denoted as $\boldsymbol{X}_{\text{mc_in}}$ and $\boldsymbol{X}_{\text{ms_in}}$, respectively. For the channel branch, the channel dimension is regarded as a spectral sequence, enabling Mamba to model long-range interchannel dependencies that are difficult to capture using conventional convolutional operations. For the spatial branch, the P^2 spatial positions within each patch are regarded as a spatial sequence, allowing the network to capture contextual relationships among spatially distributed structures. To enhance sequence modeling capability, bidirectional Mamba pathways are introduced in both branches and subsequently aggregated through learnable fusion mechanisms.

The bidirectional fusion output of the channel Mamba branch can be expressed as

$$\boldsymbol{X}_{\text{mc}} = \alpha_c \mathcal{M}_f\left(\boldsymbol{X}_{\text{mc_in}}\right) + \left(1-\alpha_c\right)\mathcal{M}_b\left(\boldsymbol{X}_{\text{mc_in}}\right). \tag{6}$$

Subsequently, statistical features are extracted, and the final output of the channel Mamba is calculated, with the unified expression

$$\boldsymbol{X}_{\text{mc_out}} = \boldsymbol{X}_{\text{m_in}} \otimes \text{MLP}\left(\text{cat}\left(\text{Std}_{seq}\left(\boldsymbol{X}_{\text{mc}}\right), \text{Mean}_{seq}\left(\boldsymbol{X}_{\text{mc}}\right)\right)\right). \tag{7}$$

Unlike the channel branch, the spatial branch additionally preserves the original spatial sequence and performs adaptive bidirectional aggregation. This design enables the model to jointly exploit local spatial structures and long-range contextual information captured by forward and backward Mamba pathways

$$\boldsymbol{X}_{\text{ms}} = \beta_1 \boldsymbol{X}_{\text{ms_in}} + \beta_2 \mathcal{M}_f\left(\boldsymbol{X}_{\text{ms_in}}\right) + \beta_3 \mathcal{M}_b\left(\boldsymbol{X}_{\text{ms_in}}\right). \tag{8}$$

After reshaping $\boldsymbol{X}_{\text{ms}}$ back to $\mathbb{R}^{N\times C_f\times P\times P}$, the spatial-enhanced output is denoted as $\boldsymbol{X}_{\text{ms_out}}$. The outputs of the channel and spatial Mamba branches are fused through feature concatenation followed by a learnable convolution

$$\boldsymbol{X}_{\text{em}} = \text{Conv}^{m}_{\text{2-D}}\left(\text{cat}\left(\boldsymbol{X}_{\text{mc_out}}, \boldsymbol{X}_{\text{ms_out}}\right)\right). \tag{9}$$

The resulting feature $\boldsymbol{X}_{\text{em}}$ integrates complementary channel-wise spectral dependencies and spatial contextual relationships. By jointly modeling spectral sequences and spatial structures through bidirectional Mamba pathways, EM enhances subtle feature variations among spectrally similar categories and provides enriched spectral and spatial representations for subsequent grouped cascaded feature modeling.

Here, $\mathcal{M}_f$ and $\mathcal{M}_b$ denote the forward and backward Mamba operators, respectively. $\alpha_c = \sigma(\theta_{\alpha_c})$ is a learnable channel-fusion coefficient that adaptively balances the contributions of the bidirectional channel Mamba pathways. $[\beta_1, \beta_2, \beta_3]$ represent normalized spatial fusion weights generated by a Softmax operation, satisfying $\sum_{i=1}^{3}\beta_i = 1$. Mean_{seq} and Std_{seq} denote sequence-wise mean and standard-deviation pooling operations, respectively. MLP denotes a multilayer perceptron, and $\text{Conv}^{m}_{\text{2-D}}$ denotes the fusion convolution used to integrate channel and spatial Mamba features.

B. Grouped Cascaded Mamba Enhancement

1) Grouped Cascaded Mamba Fusion: Building upon EM, the proposed GCMF module further strengthens local–global feature interaction, as illustrated in Fig. 2(b). GCMF combines group-wise EM-based modeling, a parallel global contextual branch, and progressive cascaded propagation to capture complementary local discriminative information and global spectral–spatial context.

Specifically, feature channels are partitioned into multiple groups for localized EM-based spectral–spatial modeling, while a global EM-enhanced feature is generated in parallel to preserve holistic contextual information. The grouped features are then progressively aggregated through cascaded concatenation, enabling effective interaction between local discriminative representations and global spectral–spatial context.

Let the input feature be $\mathbf{X}^{in}_{\text{gc}} \in \mathbb{R}^{N\times C_f\times P\times P}$, where C_f denotes the number of feature channels. The input feature is first split along the channel dimension

$$\left\{\mathbf{X}_g\right\}_{g=1}^{G} = \text{Split}_c\left(\mathbf{X}^{in}_{\text{gc}}, G\right), \quad \mathbf{X}_g \in \mathbb{R}^{N\times \frac{C_f}{G}\times P\times P}. \tag{10}$$

Subsequently, a global contextual feature $\mathbf{X}_{glb}$ is generated as

$$\mathbf{X}_{glb} = \text{EM}\left(\text{Conv}^{g}_{\text{2-D}}\left(\mathbf{X}^{in}_{\text{gc}}\right)\right). \tag{11}$$

Each grouped feature is then enhanced by EM

$$\mathbf{H}_g = \text{EM}\left(\mathbf{X}_g\right), \quad g = 1, \ldots, G. \tag{12}$$

The cross-group cascade (GC) fusion progressively integrates the global contextual feature with group-wise enhanced features

$$\mathbf{C}_g = \begin{cases} \text{cat}\left(\mathbf{H}_1, \mathbf{X}_{glb}\right), & g = 1 \\ \text{cat}\left(\mathbf{C}_{g-1}, \mathbf{H}_g\right), & 1 < g \le G. \end{cases} \tag{13}$$

The final GCMF output is obtained by projecting the cascaded representation

$$\mathbf{X}_{\text{gc}} = \text{Conv}^{o}_{\text{2-D}}\left(\mathbf{C}_G\right). \tag{14}$$

The resulting feature $\mathbf{X}_{\text{gc}}$ integrates grouped local spectral–spatial representations and global contextual information through progressive cascade fusion. By enabling hierarchical interaction among channel groups, GCMF enhances complementary feature aggregation while alleviating

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redundant channel interference, thereby providing enriched representations for subsequent spectral-guided Mamba refinement.

Here, Split_c denotes channel-wise feature partitioning, G denotes the number of channel groups, and $\mathbf{X}_g$ and $\mathbf{H}_g$ represent the grouped feature and its corresponding EM-enhanced representation, respectively. $\mathbf{X}_{glb}$ denotes the global contextual feature generated by the parallel EM branch. $\mathbf{C}_g$ represents the cascaded feature at the gth fusion stage. $\text{Conv}^g_{2\text{-D}}$ and $\text{Conv}^o_{2\text{-D}}$ denote learnable convolution mappings for global feature generation and output projection, respectively.

2) Spectral-Guided Mamba Refinement: Existing spatial Mamba modules mainly focus on modeling long-range spatial dependencies while neglecting spectral guidance, resulting in insufficient sensitivity to subtle texture variations among spectrally similar categories. In addition, independent spectral and spatial modeling may lead to feature inconsistency and limit complementary information interaction.

To further refine the GCMF-enhanced representation, a spectral-guided Mamba refinement module is introduced, as illustrated in Fig. 4. The refinement module establishes cross-guidance between channel and spatial representations, allowing channel-derived spectral cues to adaptively guide spatial sequence modeling and enhance subtle texture discrimination.

Let the input feature of the refinement module be $\mathbf{X}^{in}_{\text{em}} \in \mathbb{R}^{N\times C_f\times P\times P}$, which corresponds to the GCMF-enhanced representation in the overall pipeline.

The input feature is first projected by depthwise convolution to enhance local spatial perception

$$\mathbf{X}_m = \text{Conv}_{2\text{-D}}\left(\text{DConv}_{2\text{-D}}\left(\mathbf{X}^{in}_{\text{em}}\right)\right). \tag{15}$$

Then, dual-branch Mamba modeling is performed as

$$\begin{aligned}\mathbf{X}_c &= \text{ChaMamba}\left(\mathbf{X}_m\right)\\ \mathbf{X}_s &= \text{SpaMamba}\left(\mathbf{X}_m\right).\end{aligned} \tag{16}$$

To facilitate bidirectional interaction between spectral and spatial representations, cross-guidance gates are generated from the corresponding Mamba branches

$$\begin{aligned}\mathbf{G}_s &= \sigma\left(\phi_s\left(\mathbf{X}_s\right)\right)\\ \mathbf{G}_c &= \sigma\left(\phi_c\left(\mathbf{X}_c\right)\right)\\ \mathbf{X}_{cross} &= \mathbf{X}_c \odot \mathbf{G}_s + \mathbf{X}_s \odot \mathbf{G}_c.\end{aligned} \tag{17}$$

The cross-fused feature is further transformed into a normalized fusion representation

$$\mathbf{X}_{fu} = \text{Norm}\left(\mathcal{R}\left(\text{Conv}_{1\text{-D}}\left(\mathbf{X}_{cross}\right)\right)\right). \tag{18}$$

The enhancement weight is subsequently generated and applied to the input feature

$$\begin{aligned}\mathbf{W}_{enh} &= \text{Softmax}\left(\sigma\left(\text{Mean}\left(\mathbf{X}_{fu}\right)\right)\right)\\ \mathbf{X}_{\text{em}} &= \mathbf{X}^{in}_{\text{em}} \otimes \mathbf{W}_{enh}.\end{aligned} \tag{19}$$

The resulting feature $\mathbf{X}_{\text{em}}$ integrates spectral guidance and long-range spatial dependency modeling through bidirectional Mamba interaction. By adaptively injecting channel-aware spectral cues into spatial sequence learning, the proposed refinement module enhances subtle texture discrimination while preserving spatial structural consistency, thereby generating more discriminative representations for subsequent feature separation.

Here, $\text{DConv}_{2\text{-D}}$ denotes depthwise convolution, while ChaMamba and SpaMamba represent the channel and spatial Mamba branches, respectively. ϕ_s and ϕ_c denote learnable projection mappings for generating the guidance gates $\mathbf{G}_s$ and $\mathbf{G}_c$.

C. Cross-Feature Separation Transformer

Given the input feature $\mathbf{X}_f \in \mathbb{R}^{N\times C_f\times P\times P}$, CFST is designed to reduce the influence of redundant spectral and spatial responses and enhance complementary information through matrix-guided feature separation and cross-feature interaction. In this work, "feature separation" does not refer to strict feature disentanglement or orthogonal decomposition, but to dimension-specific restructuring of spectral and spatial representations according to their respective correlation patterns. MFS constructs spectral and spatial adjacency matrices to obtain dimension-specific separated representations, while CFT performs bidirectional interaction between the separated spectral and spatial representations. The spectral adjacency matrix (Spe-AM) $\mathbf{A}_{spe} \in \mathbb{R}^{N\times C_f\times C_f}$ models interchannel correlations, while the spatial adjacency matrix (Spa-AM) $\mathbf{A}_{spa} \in \mathbb{R}^{N\times P^2\times P^2}$ captures correlations among spatial positions within each patch. The computational complexities of spectral and spatial adjacency construction are approximately $O(NC_f^2P^2)$ and $O(NP^4C_f)$, respectively. Since both adjacency matrices are constructed within local patches rather than the entire image, the computational cost remains controllable for the adopted patch sizes.

1) Spectral Adjacency Matrix: Most existing Spe-AMs are constructed using fixed similarity rules, which limits their ability to adaptively capture dynamic correlations among spectral channels. To address this issue, the Spe-AM is constructed by jointly modeling local spectral fluctuations, global fluctuation correlations, and spectral similarity.

For notational simplicity, the batch index is omitted in the following equations. Let $\mathbf{v}_i$ denote the vectorized feature of the ith channel, and define the first-order spectral difference as

$$\mathbf{v}_i = \text{vec}\left(\mathbf{X}_f\left(i,:,:\right)\right), \quad \Delta\mathbf{v}_i = \mathbf{v}_i - \mathbf{v}_{i-1}, \quad i = 2,\ldots,C_f. \tag{20}$$

Based on the first-order spectral differences, the local fluctuation correlation and the global statistical correlation between spectral channels are calculated as

$$\begin{aligned}S_l\left(i,j\right) &= \frac{\left\|\Delta\mathbf{v}_i \odot \Delta\mathbf{v}_j\right\|_1}{\left\|\Delta\mathbf{v}_i\right\|_2 \left\|\Delta\mathbf{v}_j\right\|_2 + \epsilon}\\ S_g\left(i,j\right) &= \frac{\text{Cov}\left(\mathbf{v}_i,\mathbf{v}_j\right)}{\text{Std}\left(\mathbf{v}_i\right)\text{Std}\left(\mathbf{v}_j\right) + \epsilon}.\end{aligned} \tag{21}$$

The fluctuation correlation is then adaptively fused, and the cosine similarity is used to measure the direct spectral similarity between channel features

$$S_{flu}\left(i,j\right) = \lambda S_l\left(i,j\right) + \left(1-\lambda\right)S_g\left(i,j\right)$$

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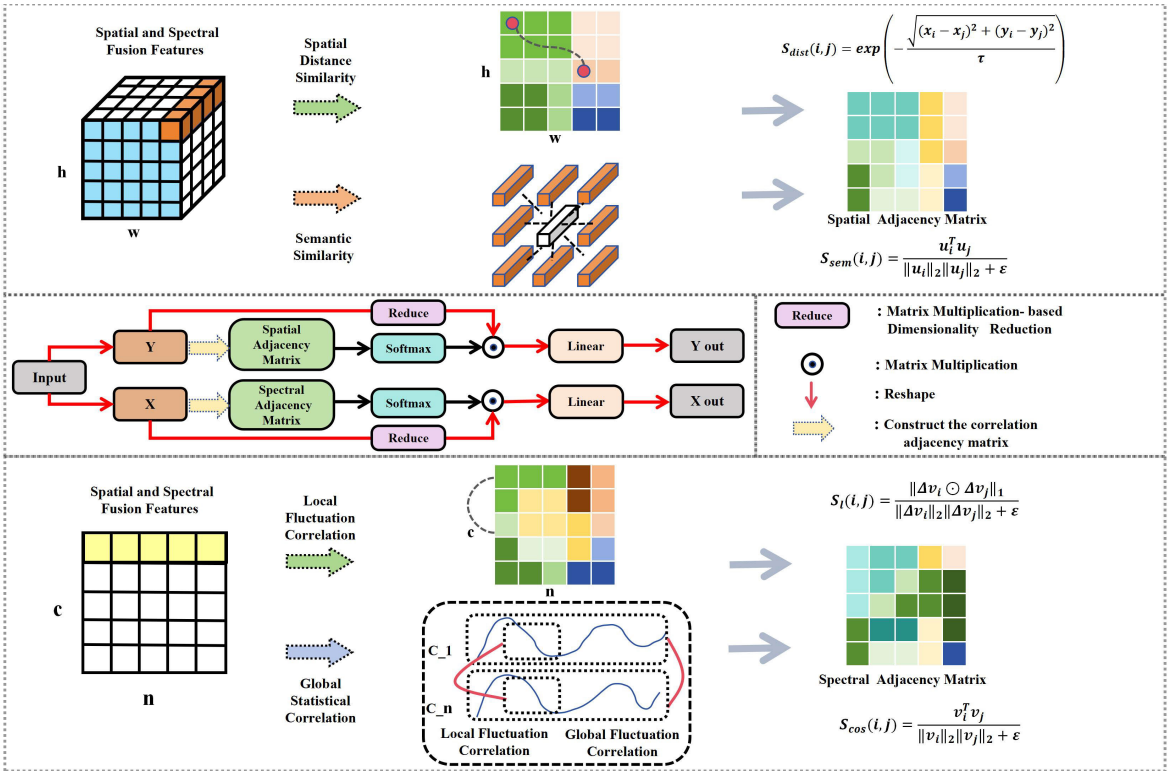


Fig. 5. MFS module. It uses spectral and spatial adjacency matrices to capture channel correlations and spatial position relationships, supporting dimension-specific feature separation and reducing the influence of redundant responses in spectrally similar scenarios.

$$S_{cos}(i,j) = \frac{\mathbf{v}_i^T \mathbf{v}_j}{\|\mathbf{v}_i\|_2 \|\mathbf{v}_j\|_2 + \epsilon}. \tag{22}$$

Finally, the Spe-AM is obtained by adaptively combining the fluctuation correlation and cosine similarity. The learnable coefficient μ controls their relative contributions, and row-wise Softmax normalization is applied to generate normalized channel-wise adjacency weights

$$\mathbf{A}_{spe}(i,j) = \mathrm{Softmax}_j\left(\mu S_{flu}(i,j) + (1-\mu) S_{cos}(i,j)\right). \tag{23}$$

Here, $\lambda = \sigma(\theta_\lambda)$ and $\mu = \sigma(\theta_\mu)$ are learnable weighting coefficients constrained to $(0,1)$. $S_{flu}(i,j)$ denotes the adaptively fused fluctuation correlation between the ith and jth spectral channels, while $S_{cos}(i,j)$ denotes their cosine similarity. ϵ is a small constant used to avoid numerical instability, and $\mathrm{Softmax}_j$ denotes row-wise normalization along the jth dimension.

2) Spatial Adjacency Matrix: Most existing spatial adjacency matrices are mainly constructed according to Euclidean distance, which neglects semantic correlations among pixels and weakens the independence of spatial feature representations. To address this limitation, the Spa-AM is constructed by jointly exploiting spatial positional proximity and pixel-level semantic similarity.

With the batch index omitted, each spatial position in the $P \times P$ patch is indexed by $i = 1, \ldots, P^2$. Let $\mathbf{u}_i \in \mathbb{R}^{C_f}$ denote the feature vector at the ith spatial position, and let (x_i, y_i) represent its corresponding spatial coordinate. The distance-based similarity between positions i and j is defined as

$$S_{dist}(i,j) = \exp\left(-\frac{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}}{\tau}\right). \tag{24}$$

The semantic similarity between positions i and j is defined as

$$S_{sem}(i,j) = \frac{\mathbf{u}_i^T \mathbf{u}_j}{\|\mathbf{u}_i\|_2 \|\mathbf{u}_j\|_2 + \epsilon}. \tag{25}$$

The final Spa-AM is obtained by adaptive fusion

$$\mathbf{A}_{spa}(i,j) = \mathrm{Softmax}_j\left(\nu S_{dist}(i,j) + (1-\nu) S_{sem}(i,j)\right). \tag{26}$$

Here, $\nu = \sigma(\theta_\nu)$ is a learnable weighting coefficient constrained to $(0,1)$, τ is a temperature coefficient, and $\mathrm{Softmax}_j$ denotes row-wise normalization along the jth dimension. The structure of MFS is shown in Fig. 5.

3) Cross-Feature Transformer: Although spectral and spatial adjacency matrices provide dimension-specific separated representations based on their respective correlation patterns, simple concatenation or weighted summation cannot fully exploit bidirectional interactions between spectral and spatial features. To address this issue, a CFT is designed, as illustrated in Fig. 2(c), to perform deep spectral–spatial interaction through dual-branch cross-attention.

After arranging the input feature along the spectral-channel and spatial-token dimensions, the corresponding representations are denoted as $\mathbf{Z}_{spe} \in \mathbb{R}^{N \times C_f \times P^2}$ and $\mathbf{Z}_{spa} \in \mathbb{R}^{N \times P^2 \times C_f}$,

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respectively. The adjacency-guided separated features are obtained as

$$\begin{aligned}\mathbf{X}_{spe} &= \mathbf{A}_{spe}\mathbf{Z}_{spe}\\ \mathbf{X}_{spa} &= \mathbf{A}_{spa}\mathbf{Z}_{spa}.\end{aligned} \tag{27}$$

Rather than simply reweighting individual responses, the spectral and spatial adjacency matrices reorganize the input features according to their respective correlation patterns, producing separated spectral and spatial representations. Correlated responses within each dimension are aggregated rather than retained as independent responses, which reduces the influence of redundant information. The subsequent CFT further exchanges complementary information between the two separated representations through bidirectional cross-feature interaction. The query, key, and value features are generated by linear projections

$$\begin{aligned}\left[\mathbf{Q}_{spe}, \mathbf{K}_{spe}, \mathbf{V}_{spe}\right] &= \text{Linear}_{spe}\left(\mathbf{X}_{spe}\right)\\ \left[\mathbf{Q}_{spa}, \mathbf{K}_{spa}, \mathbf{V}_{spa}\right] &= \text{Linear}_{spa}\left(\mathbf{X}_{spa}\right).\end{aligned} \tag{28}$$

Based on the projected query, key, and value features, bidirectional cross-attention is performed to exchange information between the spectral and spatial branches

$$\begin{aligned}\mathbf{T}_{spe} &= \text{Softmax}\left(\frac{\mathbf{Q}_{spe}\mathbf{K}_{spa}^{T}}{\sqrt{d_k}}\right)\mathbf{V}_{spa}\\ \mathbf{T}_{spa} &= \text{Softmax}\left(\frac{\mathbf{Q}_{spa}\mathbf{K}_{spe}^{T}}{\sqrt{d_k}}\right)\mathbf{V}_{spe}.\end{aligned} \tag{29}$$

The two cross-attention responses are further dimension-aligned and concatenated to generate the final CFT output

$$\mathbf{X}_{\text{cft}} = \text{MLP}\left(\text{cat}\left(\text{Proj}_{spe}\left(\mathbf{T}_{spe}\right), \text{Proj}_{spa}\left(\mathbf{T}_{spa}\right)\right)\right). \tag{30}$$

The resulting feature $\mathbf{X}_{cft}$ integrates spectral-guided spatial responses and spatial-guided spectral responses. By combining adjacency-aware feature separation with bidirectional cross-attention fusion, CFT explicitly enhances interactions between spectral and spatial representations, thereby improving the discriminability of fused features for spectrally similar categories.

Here, $\mathbf{Z}_{spe}$ and $\mathbf{Z}_{spa}$ denote the spectral-channel and spatial-token arranged representations, respectively. d_k denotes the key dimension used for attention scaling. $\mathbf{T}_{spe}$ and $\mathbf{T}_{spa}$ represent spectral-guided spatial aggregation and spatial-guided spectral aggregation, respectively. Proj_{spe} and Proj_{spa} denote projection mappings used to align spectral and spatial features before concatenation.

D. Algorithm Summarization for GCEMNet

The overall process of the proposed GCEMNet is summarized in Algorithm 1.

Algorithm 1 Overall Procedure of GCEMNet

Require: Hyperspectral image $\mathbf{I} \in \mathbb{R}^{H\times W\times B}$; retained PCA dimension C; patch size P; number of channel groups G.

Ensure: Predicted label $\hat{\mathbf{y}}$ for each input patch.

Preprocessing
Apply PCA to reduce spectral dimensionality:
  $\mathbf{I}_{pca} \leftarrow \text{PCA}(\mathbf{I}, C)$, where $\mathbf{I}_{pca} \in \mathbb{R}^{H\times W\times C}$.
Extract local spatial patches from $\mathbf{I}_{pca}$:
  $\mathbf{X} \leftarrow \text{PatchExtract}(\mathbf{I}_{pca}, P)$, where $\mathbf{X} \in \mathbb{R}^{N\times C\times P\times P}$.
Normalize input patches:
  $\mathbf{X}_{in} \leftarrow \text{Norm}(\mathbf{X})$.
Spectral–Spatial Fusion Extraction
Extract shallow spectral and spatial features:
  $\mathbf{F}_0 \leftarrow \text{Conv}(\mathbf{X}_{in})$.
Apply channel–spatial attention fusion:
  $\mathbf{F}_{csaf} \leftarrow \text{CSAF}(\mathbf{F}_0)$.
Apply enhanced Mamba fusion:
  $\mathbf{F}_{ref}^{0} \leftarrow \text{EM}(\mathbf{F}_{csaf})$.
Grouped Cascaded Mamba Enhancement
Perform grouped cascaded Mamba fusion:
  $\mathbf{F}_{gc} \leftarrow \text{GCMF}(\mathbf{F}_{ref}^{0}, G)$.
Apply spectral-guided Mamba refinement:
  $\mathbf{F}_{ref} \leftarrow \text{Refine}(\mathbf{F}_{gc})$.
Matrix Feature Separation
Construct the spectral adjacency matrix:
  $\mathbf{A}_{spe} \leftarrow \text{SpeAM}(\mathbf{F}_{ref})$.
Construct the spatial adjacency matrix:
  $\mathbf{A}_{spa} \leftarrow \text{SpaAM}(\mathbf{F}_{ref})$.
Obtain adjacency-separated spectral and spatial features:
  $(\mathbf{F}_{spe}, \mathbf{F}_{spa}) \leftarrow \text{MFS}(\mathbf{F}_{ref}, \mathbf{A}_{spe}, \mathbf{A}_{spa})$.
Cross-Feature Fusion and Classification
Perform bidirectional cross-feature Transformer fusion:
  $\mathbf{F}_{cft} \leftarrow \text{CFT}(\mathbf{F}_{spe}, \mathbf{F}_{spa})$.
Predict the class label:
  $\hat{\mathbf{y}} \leftarrow \text{Classifier}(\mathbf{F}_{cft})$.
return: $\hat{\mathbf{y}}$

IV. Experiments

This section evaluates the proposed GCEMNet from six aspects: comparison with representative HSI classification methods, parameter sensitivity analysis, module replacement analysis, ablation study, computational complexity evaluation, and visualization analysis. The compared methods include traditional machine learning classifiers and recent deep learning models for HSI classification. Experiments are conducted on two mineral hyperspectral datasets and one agricultural hyperspectral dataset, covering spectrally similar mineral categories, fine-grained crop classes, and imbalanced sample distributions. These datasets provide challenging benchmarks for evaluating the classification accuracy, module effectiveness, and computational cost of the proposed method under spectrally similar scenarios.

A. Datasets

We conducted comprehensive comparative experiments with state-of-the-art methods on two challenging benchmark datasets for mineral HSI classification: the Holden Crater dataset and the Nili Fossae dataset [41], which contain 6 and 9 distinct mineral classes, respectively. These datasets pose significant classification challenges due to the high interclass

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TABLE I

DETAILED INFORMATION OF HOLDEN CRATER AND CLASSIFICATION RESULTS UNDER DIFFERENT PCA DIMENSIONS

class No.	Class Name	Sample Size		Result (%)			
		Training	Testing	PCA10	PCA15	PCA30	PCA50
1	Analcime	9	931	**99.32 ± 0.34**	99.15 ± 0.75	98.56 ± 1.35	98.34 ± 0.60
2	Plagioclase	15	1457	93.95 ± 2.30	**97.68 ± 1.43**	93.93 ± 0.96	93.66 ± 1.88
3	Prehnite	15	1545	**94.42 ± 2.41**	93.90 ± 3.47	94.11 ± 2.77	93.38 ± 2.14
4	High-Ca Pyroxene	69	6894	99.96 ± 0.06	**99.97 ± 0.15**	99.65 ± 0.19	98.82 ± 1.05
5	Serpentine	87	8610	99.92 ± 0.08	**99.97 ± 0.03**	99.89 ± 0.10	99.77 ± 0.21
6	Margarite	5	453	96.29 ± 1.58	**96.34 ± 1.24**	92.36 ± 3.78	77.47 ± 11.80
OA%				98.96 ± 0.09	**99.21 ± 0.17**	98.69 ± 0.13	97.92 ± 0.51
AA%				97.31 ± 0.32	**97.83 ± 0.46**	96.42 ± 0.69	93.57 ± 2.10
$\kappa \times 100$				98.46 ± 0.13	**98.75 ± 0.26**	98.22 ± 0.25	96.93 ± 0.76

TABLE II

DETAILED INFORMATION OF NILI FOSSAE AND CLASSIFICATION RESULTS UNDER DIFFERENT PCA DIMENSIONS

class No.	Class Name	Sample Size		Result (%)			
		Training	Testing	PCA3	PCA5	PCA10	PCA15
1	Fe-Olivine	85	8478	99.28 ± 0.45	99.50 ± 0.51	**99.63 ± 0.21**	99.48 ± 0.70
2	Epidote	10	986	52.92 ± 12.80	73.73 ± 5.70	83.07 ± 9.35	**87.24 ± 6.90**
3	Chlorite	4	382	**89.48 ± 3.72**	81.55 ± 8.00	66.86 ± 9.10	58.72 ± 12.05
4	Bassanite	33	3253	95.52 ± 2.34	94.63 ± 1.40	96.19 ± 1.0	**96.39 ± 0.66**
5	Illite/Muscovite	2	226	29.65 ± 7.30	**46.28 ± 11.43**	42.48 ± 6.68	33.01 ± 4.09
6	Mg-Carbonate	49	4818	**98.96 ± 0.36**	98.51 ± 0.93	98.12 ± 1.91	97.54 ± 1.14
7	Plagioclase	10	1047	79.91 ± 3.56	**94.24 ± 3.06**	90.39 ± 4.70	86.18 ± 2.76
8	Prehnite	12	1149	**91.18 ± 7.00**	90.53 ± 2.96	84.95 ± 3.84	85.20 ± 3.33
9	Serpentine	62	6104	**97.44 ± 0.42**	96.51 ± 0.53	96.86 ± 0.77	96.47 ± 0.60
OA%				94.75 ± 0.57	**95.76 ± 0.34**	95.71 ± 0.49	95.29 ± 0.45
AA%				81.59 ± 1.81	**86.16 ± 2.03**	84.28 ± 2.19	82.25 ± 1.86
$\kappa \times 100$				93.35 ± 0.73	**94.97 ± 0.44**	94.56 ± 0.63	94.03 ± 0.56

TABLE III

DETAILED INFORMATION OF WHU-HI-HANCHUAN AND CLASSIFICATION RESULTS UNDER DIFFERENT PCA DIMENSIONS

class No.	Class Name	Sample Size		Result (%)			
		Training	Testing	PCA10	PCA15	PCA30	PCA50
1	Strawberry	456	44279	99.83 ± 0.13	99.84 ± 0.05	**99.86 ± 0.04**	99.84 ± 0.07
2	Cowpea	231	22522	98.76 ± 1.31	**99.01 ± 0.43**	98.98 ± 0.28	98.96 ± 0.43
3	Soybean	105	10182	99.70 ± 0.16	99.79 ± 0.10	99.60 ± 0.20	**99.84 ± 0.08**
4	Sorghum	54	5299	**99.68 ± 0.08**	99.60 ± 0.13	99.55 ± 0.14	99.65 ± 0.14
5	Water spinach	12	1188	**98.44 ± 0.89**	97.55 ± 1.31	97.84 ± 0.88	97.90 ± 1.35
6	Watermelon	46	4487	91.94 ± 1.55	90.04 ± 2.60	**92.93 ± 1.16**	91.05 ± 1.10
7	Greens	59	5844	**94.76 ± 0.88**	93.99 ± 0.44	93.98 ± 0.53	94.50 ± 0.80
8	Trees	183	17795	97.03 ± 1.74	97.64 ± 0.22	**97.83 ± 0.29**	97.54 ± 0.28
9	Grass	95	9374	98.30 ± 0.35	98.44 ± 0.51	**98.64 ± 0.25**	98.31 ± 0.43
10	Red roof	106	10410	98.77 ± 0.50	98.62 ± 0.24	**98.90 ± 0.20**	98.65 ± 0.21
11	Gray roof	174	16737	98.94 ± 0.27	99.16 ± 0.18	**99.25 ± 0.18**	99.22 ± 0.33
12	Plastic	39	3640	99.54 ± 0.52	**99.71 ± 0.24**	99.07 ± 0.58	99.38 ± 0.77
13	Bare soil	91	9025	91.23 ± 0.96	90.91 ± 1.13	**91.71 ± 0.84**	91.04 ± 1.57
14	Road	188	18372	**99.09 ± 0.42**	98.73 ± 0.61	98.22 ± 0.36	98.77 ± 0.82
15	Bright object	11	1125	79.53 ± 5.37	81.92 ± 1.94	**82.33 ± 2.41**	81.56 ± 2.58
16	Water	768	74633	98.81 ± 3.45	**99.97 ± 0.02**	99.95 ± 0.04	99.97 ± 0.03
OA%				98.36 ± 1.28	98.71 ± 0.08	**98.76 ± 0.04**	98.73 ± 0.14
AA%				96.52 ± 0.75	96.56 ± 0.22	**96.79 ± 0.25**	96.64 ± 0.31
$\kappa \times 100$				98.09 ± 1.48	98.48 ± 0.09	**98.57 ± 0.09**	98.51 ± 0.16

similarity in spatial patterns and spectral features across different mineral regions. To further evaluate the applicability of the proposed method in agricultural scenarios, we include the WHU-Hi-HanChuan hyperspectral dataset [42], a real-world agricultural benchmark containing 16 crop classes with similar spectral–spatial characteristics. This setting provides additional evidence for evaluating GCEMNet in both geological and agricultural scenes. In addition, all three datasets contain imbalanced class distributions, which allows us to examine the behavior of different methods on minority classes. The class-wise training and testing sample numbers are listed together with the PCA sensitivity results in Tables I–III.

B. Experimental Setup

1) Evaluation Metrics: We evaluate all methods using three commonly used metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). OA measures the overall classification accuracy, AA reflects the average class-wise accuracy and is more sensitive to minority classes, and κ evaluates the agreement between predicted and reference labels. Higher values indicate better classification performance.

2) Configuration: All experiments were conducted on a workstation equipped with a 13th Gen Intel[1] Core[2] i9-13900K CPU and dual NVIDIA GeForce RTX 3090 24GB GPUs (Operating System: Linux 5.15.0-107-generic, CUDA Version: 12.4, Driver Version: 550.90.07).

3) Experimental Details: We validated the effectiveness of the proposed method against traditional machine learning-based models and deep learning models for HSI classification, including SVM [13], RF [14], PyResNet [43], ContextualNet [44], SSRN [45], SSFTT [46], MATNet [47], SS-MSDCNet [48], CSCANet [49], MambaHsi [50], HsiMamba+ [51], and DBMGNet [52]. For fair comparison, all methods are evaluated under the same training and testing sample settings for each dataset. The results of RF and all deep learning models are reported as the mean and standard deviation over ten independent runs to reduce the influence of random initialization and training fluctuations. For GCEMNet, the batch size is set to 64, PCA is used for spectral dimensionality reduction, and the number of training epochs is set to 200 on all three datasets. According to the parameter sensitivity analyses, the spectral and Spa-AM configurations $(K_{\text{spe}}, K_{\text{spa}})$ are set to $(4, 4)$ for Holden Crater, $(8, 8)$ for Nili Fossae, and $(16, 32)$ for WHU-Hi-HanChuan, respectively.

C. Parameter Settings Analysis

To investigate the influence of key parameters, we conduct sensitivity analyses on PCA dimensionality, patch size, group number, and adjacency matrix configuration. In each analysis, only the target parameter is varied while the other settings are kept fixed. These experiments examine how spectral representation, spatial receptive field, grouped sequence modeling, and spectral–spatial adjacency modeling affect GCEMNet on different datasets.

1) Pca Dimension Reduction: HSIs usually contain high-dimensional spectra with redundant and noisy bands, increasing computational cost and weakening the discrimination of spectrally similar ground objects. PCA is therefore employed to reduce spectral dimensionality while preserving the major spectral information. We conduct sensitivity analysis on three datasets to investigate the effect of PCA dimensionality on GCEMNet. The results show that the optimal PCA dimension varies with scene-specific spectral complexity, noise level, and class distribution.

For the Holden Crater dataset, the best overall performance is achieved with 15 PCA components. Lower dimensions may discard weak but discriminative mineral absorption features, while higher dimensions retain redundant components and noise, degrading the recognition of small-sample classes such as Margarite.

For the Nili Fossae dataset, a compact representation with 5 PCA components achieves the best OA, AA, and κ. Lower dimensions fail to preserve identifiable mineral cues, whereas higher dimensions introduce redundant spectral information

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and increase confusion among minerals with similar spectral responses.

For the WHU-Hi-HanChuan dataset, 30 PCA components provide the best overall performance. The diverse crop types and complex spatial distributions require richer spectral representations. Lower dimensions may discard fine-grained crop-discriminative features, whereas excessively high dimensions introduce redundant spectral noise. Therefore, 30 PCA components provide a balanced representation for this dataset.

These results indicate that PCA dimensionality should be selected according to the spectral complexity of each scene. Too few components may remove weak but useful spectral cues, whereas too many components may retain redundant or noisy bands. The adopted PCA settings therefore reflect a tradeoff between information preservation and redundancy suppression. Detailed sample statistics and quantitative results are reported in Tables I–III.

2) Patch Size: Patch size determines the spatial receptive field for local spectral–spatial feature extraction. A small patch may provide insufficient spatial context, whereas an overly large patch may cross category boundaries and introduce background interference. Therefore, we conduct sensitivity analysis with different patch sizes to examine the influence of spatial context on classification performance.

For the Holden Crater dataset, minerals are regionally aggregated with moderate spatial scales. Small patches provide limited neighborhood information, whereas large patches may cover adjacent mineral regions and introduce mixed spatial responses. The 15 $\times$ 15 patch achieves the best performance by balancing contextual completeness and feature purity.

For the Nili Fossae dataset, mineral distributions are more fragmented and affected by stronger surface interference. Larger patches, such as 11 and 13, may introduce more background noise and reduce OA and AA, while smaller patches provide insufficient local representation. The compact 9 $\times$ 9 patch focuses more effectively on local mineral characteristics.

For the WHU-Hi-HanChuan dataset, crop fields usually occupy larger contiguous regions. Smaller patch sizes, such as 17 and 19, may not fully capture field-level spatial patterns, whereas an overly large patch, such as 25, may introduce interference from roads, buildings, and bare soil. The 23 $\times$ 23 patch provides the best balance between crop-region coverage and background suppression.

The results show that both insufficient and excessive spatial context can reduce classification accuracy. The detailed patch size comparison is presented in Table IV.

3) Number of Groups: To analyze the influence of the group number on the grouped Mamba enhancement module, we conduct a sensitivity analysis while keeping other parameters fixed. The results indicate that the appropriate group number varies with spectral complexity, sequence length, and class distribution.

The comparison between the nongrouped and noncascaded schemes in Table V shows relatively small performance differences, indicating that grouping alone provides limited performance gains. The improvement becomes more evident when grouped feature modeling is combined with cascaded propagation. Grouping facilitates fine-grained local feature modeling by partitioning long spectral–spatial sequences, while the cascaded mechanism progressively propagates complementary information across groups. These two designs therefore play complementary roles in improving the classification capability of the GCMF module.

TABLE IV

CLASSIFICATION RESULTS OF THREE DATASETS WITH DIFFERENT PATCH SIZES

Dataset	Patch Size	OA (%)	AA (%)	$\kappa \times 100$
Holden	9	97.83 ± 0.39	95.31 ± 1.61	96.80 ± 0.58
	11	98.39 ± 0.43	96.59 ± 1.73	97.61 ± 0.64
	13	99.02 ± 0.13	97.42 ± 0.69	98.55 ± 0.25
	15	**99.21 ± 0.17**	**97.83 ± 0.46**	**98.75 ± 0.26**
	17	98.87 ± 0.27	97.22 ± 0.72	98.32 ± 0.39
Nili	5	92.55 ± 1.10	77.47 ± 2.59	89.25 ± 1.41
	7	95.35 ± 0.50	82.32 ± 1.04	93.79 ± 0.63
	9	**95.76 ± 0.35**	**86.16 ± 2.03**	**94.97 ± 0.44**
	11	95.56 ± 0.70	85.10 ± 1.95	94.37 ± 0.88
	13	93.81 ± 0.83	78.29 ± 1.32	92.14 ± 1.05
Hanchuan	17	98.41 ± 0.10	95.97 ± 0.25	98.14 ± 0.11
	19	98.59 ± 0.08	96.36 ± 0.20	98.35 ± 0.09
	21	98.71 ± 0.09	96.69 ± 0.23	98.49 ± 0.10
	23	**98.76 ± 0.04**	**96.79 ± 0.25**	**98.57 ± 0.09**
	25	98.66 ± 0.28	96.59 ± 0.47	98.43 ± 0.33

TABLE V

CLASSIFICATION RESULTS OF THREE DATASETS WITH DIFFERENT GROUP NUMBERS

Dataset	Result	Number of Groups					
		Non-Grouped	Non-Cascaded	2	4	8	16
Holden	OA (%)	99.04 ± 0.17	99.06 ± 0.09	99.00 ± 0.30	99.10 ± 0.17	**99.21 ± 0.17**	99.03 ± 0.39
	AA (%)	97.62 ± 0.48	97.65 ± 0.38	97.47 ± 0.65	97.54 ± 0.48	**97.83 ± 0.46**	97.38 ± 0.93
	$\kappa \times 100$	98.58 ± 0.26	98.62 ± 0.16	98.53 ± 0.44	98.67 ± 0.25	**98.75 ± 0.26**	98.57 ± 0.58
Nili	OA (%)	95.18 ± 0.47	95.35 ± 0.67	**95.76 ± 0.34**	95.33 ± 0.41	95.55 ± 0.87	95.61 ± 0.58
	AA (%)	86.50 ± 1.75	**86.52 ± 1.86**	86.16 ± 2.03	85.73 ± 1.16	85.66 ± 2.35	85.71 ± 1.06
	$\kappa \times 100$	94.91 ± 0.59	94.93 ± 0.52	**94.97 ± 0.44**	94.73 ± 0.52	94.49 ± 1.10	94.72 ± 0.73
Hanchuan	OA (%)	98.19 ± 0.03	98.22 ± 0.03	98.52 ± 0.15	**98.76 ± 0.04**	98.38 ± 0.21	97.88 ± 0.10
	AA (%)	95.56 ± 0.54	95.76 ± 0.43	96.22 ± 0.41	**96.79 ± 0.25**	95.69 ± 0.42	94.58 ± 0.27
	$\kappa \times 100$	97.88 ± 0.20	97.91 ± 0.25	98.27 ± 0.14	**98.57 ± 0.09**	98.11 ± 0.17	97.53 ± 0.15

For the Holden Crater dataset, which has relatively clear mineral spectral structures and high interclass discrimination, 8 groups provide fine-grained local decoupling while preserving global spectral correlations through grouped cascade, yielding the best performance. Excessive grouping, such as 16 groups, may break spectral continuity and degrade classification accuracy.

For the Nili Fossae dataset, which contains heavy spectral interference and easily confused mineral categories, two groups maintain stable global modeling and effectively suppress noise. Too many groups may disrupt spectral consistency and reduce OA and AA.

For the WHU-Hi-HanChuan dataset, which contains numerous crop classes with subtle spectral differences, four groups strike a good balance between fine-grained feature decoupling and global band correlation. Insufficient or excessive grouping impairs feature integrity and leads to lower classification performance.

Overall, the group number controls the balance between local feature decoupling and global spectral continuity. The detailed comparison is reported in Table V.

4) Adjacency Matrix Number Analysis: To investigate the influence of the number of independently constructed adjacency matrices, we first conduct an independent sensitivity analysis for the spectral and spatial adjacency branches on all three datasets. Let K_{spe} and K_{spa} denote the numbers of spectral and spatial adjacency matrices, respectively. For the

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TABLE VI

OA RESULTS OF DIFFERENT NUMBERS OF SPECTRAL AND SPATIAL ADJACENCY MATRICES

Dataset	Branch	Number of Adjacency Matrices					
		2	4	8	16	32	64
Holden	Spectral	97.42±0.31	**97.88±0.18**	97.76±0.24	97.61±0.29	97.49±0.36	97.37±0.41
	Spatial	97.35±0.28	**97.91±0.16**	97.79±0.21	97.63±0.27	97.46±0.34	97.31±0.39
Nili	Spectral	93.71±0.65	94.08±0.52	**94.63±0.37**	94.44±0.43	94.19±0.56	93.96±0.69
	Spatial	93.55±0.72	94.02±0.57	**94.51±0.41**	94.35±0.46	94.10±0.59	93.88±0.74
Hanchuan	Spectral	95.34±0.14	96.22±0.11	96.88±0.08	**97.29±0.06**	97.10±0.09	96.74±0.13
	Spatial	95.18±0.15	96.05±0.12	96.83±0.09	97.26±0.07	**97.46±0.05**	97.12±0.10

TABLE VII

OA RESULTS OF THE CONDITIONAL SENSITIVITY ANALYSIS WITH THE COMPLEMENTARY BRANCH FIXED AT ITS PREFERRED MATRIX NUMBER

Dataset	Branch	Number of Adjacency Matrices					
		2	4	8	16	32	64
Holden	Spectral ($K_{spa}=4$)	98.94±0.21	**99.21±0.17**	99.08±0.19	98.91±0.23	98.74±0.27	98.61±0.32
	Spatial ($K_{spe}=4$)	98.90±0.20	**99.21±0.17**	99.10±0.18	98.93±0.22	98.78±0.26	98.60±0.31
Nili	Spectral ($K_{spa}=8$)	95.31±0.46	95.52±0.40	**95.76±0.34**	95.61±0.39	95.38±0.47	95.11±0.55
	Spatial ($K_{spe}=8$)	95.28±0.49	95.49±0.43	**95.76±0.34**	95.58±0.41	95.34±0.50	95.08±0.58
Hanchuan	Spectral ($K_{spa}=32$)	98.31±0.10	98.48±0.08	98.63±0.06	**98.76±0.04**	98.69±0.06	98.54±0.09
	Spatial ($K_{spe}=16$)	98.26±0.11	98.42±0.09	98.56±0.07	98.68±0.05	**98.76±0.04**	98.61±0.08

spectral adjacency analysis, K_{spa} is fixed to 1, while K_{spe} is varied among 2, 4, 8, 16, 32, and 64. Likewise, for the spatial adjacency analysis, K_{spe} is fixed to 1, while K_{spa} is varied over the same range. By changing only one branch at a time, this first-stage analysis identifies the preferred matrix number for each branch independently.

As shown in Table VI, the preferred numbers of spectral and spatial adjacency matrices vary across the three datasets. For Holden Crater, both branches achieve their highest OA when the number of adjacency matrices is set to 4. Nili Fossae favors 8 adjacency matrices for both spectral and spatial modeling. In contrast, WHU-Hi-HanChuan exhibits different pReferences for the two branches, with the spectral branch reaching its highest OA at $K_{spe} = 16$ and the spatial branch at $K_{spa} = 32$. These results indicate that increasing the number of adjacency matrices does not continuously improve classification performance. Instead, the performance generally increases until an appropriate matrix number is reached and then gradually decreases as additional matrices are introduced.

Based on this independent sensitivity analysis, the preferred spectral and spatial matrix numbers are denoted as K^*_{spe} and K^*_{spa}, respectively. The preliminary selections are therefore $(4, 4)$ for Holden Crater, $(8, 8)$ for Nili Fossae, and $(16, 32)$ for WHU-Hi-HanChuan.

To further verify whether the independently selected matrix numbers remain appropriate when the spectral and spatial branches operate jointly, we conduct a second-stage conditional sensitivity analysis. Specifically, when evaluating the spectral branch, K_{spa} is fixed to the preferred spatial matrix number K^*_{spa} obtained from the first-stage analysis, while K_{spe} is varied among 2, 4, 8, 16, 32, and 64. Conversely, when evaluating the spatial branch, K_{spe} is fixed to the preferred spectral matrix number K^*_{spe}, while K_{spa} is varied over the same range. This conditional evaluation examines whether the independently identified matrix numbers remain optimal when the complementary branch is configured at its preferred setting.

As shown in Table VII, the preferred matrix numbers identified in the first-stage independent analysis remain optimal when the complementary branch is fixed at its corresponding preferred setting. For Holden Crater, both the spectral and spatial branches achieve their highest OA at 4 adjacency matrices. For Nili Fossae, the best performance of both branches is obtained with 8 matrices. For WHU-Hi-HanChuan, the spectral branch reaches its highest OA at $K_{spe} = 16$ when K_{spa} is fixed to 32, whereas the spatial branch achieves its best result at $K_{spa} = 32$ when K_{spe} is fixed to 16.

The consistency between the independent and conditional sensitivity analyses confirms the selected adjacency matrix configurations of $(4, 4)$ for Holden Crater, $(8, 8)$ for Nili Fossae, and $(16, 32)$ for WHU-Hi-HanChuan. In particular, the asymmetric configuration on WHU-Hi-HanChuan further indicates that the preferred spectral and spatial matrix numbers are not necessarily identical, but depend on dataset-specific spectral complexity and spatial structure.

Taken together, the two-stage adjacency matrix analysis provides a systematic procedure for determining the spectral and spatial matrix numbers. The first-stage independent analysis examines the influence of each branch while fixing the complementary branch to a single adjacency matrix, thereby identifying the preferred matrix number for spectral and spatial correlation modeling separately. The second-stage conditional analysis further evaluates each branch with the complementary branch fixed at its independently selected preferred value, verifying whether the selected values remain stable under joint spectral and spatial modeling.

The results consistently support the final configurations of $(4, 4)$ for Holden Crater, $(8, 8)$ for Nili Fossae, and $(16, 32)$ for WHU-Hi-HanChuan. The nonmonotonic performance trends observed in the sensitivity analyses indicate that continuously increasing the number of adjacency matrices does not necessarily improve classification performance. Moreover, the asymmetric configuration on WHU-Hi-HanChuan demonstrates that the spectral and spatial matrix numbers should not be assumed to be strictly coupled. Instead, their appropriate values depend on the spectral and spatial characteristics of the specific dataset.

Overall, the parameter analyses of GCEMNet systematically examine several key architectural and input-related factors, including PCA dimensionality, patch size, group number, and adjacency matrix configuration. These experiments, respectively, investigate spectral representation capacity, spatial receptive fields, grouped sequence modeling, and correlation-aware spectral and spatial feature separation. Together, they cover the major factors that affect feature extraction and class discrimination in spectrally similar hyperspectral classification scenarios. Moreover, all parameter settings are quantitatively evaluated on Holden Crater, Nili Fossae, and WHU-Hi-HanChuan, which exhibit different spectral complexity, spatial structures, and class distributions. Therefore, the adopted parameter settings are supported by systematic comparisons across three representative datasets rather than being empirically selected on a single dataset.

D. Module Comparison Analysis

To further evaluate the design rationality of the proposed modules, we conduct module replacement experiments

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TABLE VIII
COMPARISON RESULTS OF DIFFERENT SPECTRAL–SPATIAL MODULES ON HOLDEN CRATER

Spectral-Spatial Module	OA (%)	AA (%)	$\kappa \times 100$
SSFE	97.89±0.46	96.01±0.27	97.28±0.48
PAM+CAM	98.11±0.11	95.92±0.35	97.21±0.31
MSC-Attn-p	98.85±0.51	96.81±0.61	98.31±0.33
CSAF (ours)	**99.21±0.17**	**97.83±0.46**	**98.75±0.26**

TABLE IX
COMPARISON RESULTS OF DIFFERENT MAMBA ENHANCEMENT MODULES ON HOLDEN CRATER

Mamba Enhancement Module	OA (%)	AA (%)	$\kappa \times 100$
GC+SA	98.35±0.63	96.34±1.11	97.54±1.44
GC+SCA	98.49±0.22	96.48±0.48	97.48±0.32
GC+BM	98.58±0.14	96.72±0.22	97.90±0.20
GC+BMplus	98.77±0.19	97.33±0.36	98.19±0.20
GC+EM (ours)	**99.21±0.17**	**97.83±0.46**	**98.75±0.26**

TABLE X
COMPARISON RESULTS OF DIFFERENT SEPARATION MODULES ON HOLDEN CRATER

Separation Module	OA (%)	AA (%)	$\kappa \times 100$
TFS	98.62±0.32	97.23±0.63	97.95±0.53
TokenLearner	98.29±0.30	95.05±0.99	97.47±0.38
CFST (ours)	**99.21±0.17**	**97.83±0.46**	**98.75±0.26**

on the Holden Crater dataset. Three types of components are compared: spectral–spatial attention modules, Mamba enhancement modules, and feature separation modules. In each comparison, only the corresponding module is replaced, while the remaining network structure and training settings are kept unchanged.

1) Channel–Spatial Attention Module: Three functionally equivalent modules of the same type were adopted for comparative verification [46], [53], [54];

2) EM Module: All grouped modules adopt the same group cascading structure with only the Mamba enhancement unit replaced. We compare the performance of similar Mamba modules and traditional attention modules [47], [49], [50], [51].

3) Feature Separation Module: The comparison was made between the TFS module using dual-branch projection [49] and the TokenLearner module using a Transformer attention mechanism [55].

Experimental results show that the proposed modules achieve the best overall performance among the compared alternatives, indicating their effectiveness under the same network framework.

In the spectral–spatial attention comparison, SSFE, PAM + CAM, and MSC-Attn-p provide useful feature enhancement but still model spectral and spatial information in a relatively separated or limited manner. SSFE mainly focuses on spectral feature extraction, PAM + CAM combines spatial and channel attention without deeper spectral–spatial interaction, and MSC-Attn-p emphasizes multiscale spectral responses. In contrast, CSAF adaptively integrates channel-wise and spatial structural cues, enabling the model to better preserve fine-grained shallow spectral and spatial features. As shown in Table VIII, CSAF achieves the best OA, AA, and κ among the compared modules.

In the Mamba enhancement comparison, traditional attention variants such as GC + SA and GC + SCA provide limited long-range dependency modeling. Existing Mamba-based variants, including GC + BM and GC + BMplus, improve sequential feature modeling but still lack sufficient feature refinement for spectrally similar categories. In contrast, the proposed EM module enhances long-range spectral–spatial dependencies while preserving discriminative local responses, leading to better overall classification performance.

In the feature separation comparison, TFS relies on dual-branch projection and may retain redundant responses, while TokenLearner performs token selection but may discard weak yet useful features in spectrally similar scenarios. CFST reduces the influence of redundant spectral and spatial responses and strengthens complementary feature interaction. As a result, it achieves better overall performance and produces more discriminative feature representations.

Detailed module replacement results are reported in Tables VIII–X.

E. Experimental Results and Visualization

Tables XI–XIII report the quantitative comparison results on the three datasets. The results of GCEMNet and the compared deep learning methods are reported as the mean and standard deviation over ten independent runs. GCEMNet achieves the best OA, AA, and κ on all datasets, showing strong classification capability in spectrally similar mineral and agricultural scenarios. Compared with traditional machine learning methods and recent deep learning models, GCEMNet consistently improves the overall metrics, demonstrating the effectiveness of shallow feature fusion, grouped cascaded Mamba enhancement, and matrix-guided feature separation under different data distributions and complex spatial patterns.

Although GCEMNet achieves the best overall performance, it does not obtain the highest accuracy for every class. For example, Epidote and Illite/Muscovite in the Nili Fossae dataset remain challenging because only 10 and 2 training samples are available, respectively. The Bright Object class in the WHU-Hi-HanChuan dataset also shows relatively lower accuracy with only 11 training samples. These cases indicate that extreme sample scarcity, class imbalance, and strong spectral overlap can still limit the learning of stable discriminative representations. Nevertheless, the best AA on all three datasets shows that GCEMNet generally improves class-wise recognition while maintaining strong overall performance.

To illustrate the improvement in interclass discriminability introduced by CFST, PCA dimensionality reduction and k-means clustering are used to visualize the 2-D feature distributions before and after CFST processing, as shown in Fig. 6. Without CFST, features from similar categories tend to overlap and show blurred boundaries. After introducing CFST, the distributions become more compact within each class and more separable across classes, indicating that matrix-guided feature separation and cross-feature interaction help reduce the

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TABLE XI

DETAILED CLASSIFICATION ACCURACY OF HOLDEN CRATER ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	CSCANet	MambaHsi	HsiMamba+	DBMGNet	ours
1	87.97	97.53±1.38	74.51±26.30	85.79±6.00	83.11±6.88	99.43±0.29	96.79±2.41	98.53±1.07	98.44±0.88	98.92 ± 0.69	99.38±0.40	**99.67±0.17**	99.15 ± 0.75
2	22.03	75.86±1.55	87.08±5.05	91.56±4.75	92.69±5.70	94.02±3.72	91.57±5.10	91.29±1.03	93.14±2.24	94.65 ± 1.05	94.92±1.86	92.26±1.43	**97.68 ± 1.43**
3	91.84	84.19±1.23	79.83±14.76	87.71±5.28	86.77±8.56	89.75±1.63	91.28±4.16	93.05±3.46	91.13±2.03	90.71 ± 0.92	91.61±1.69	**95.21±0.88**	93.90 ± 3.47
4	98.49	99.07±0.19	91.23±10.49	97.08±1.95	96.32±2.36	98.39±1.13	97.90±1.50	98.50±0.36	99.09±0.55	99.49 ± 0.28	99.63±0.23	99.48±0.51	**99.97 ± 0.15**
5	99.98	**100.00±0.00**	98.53±1.27	99.28±1.59	98.69±0.81	99.68±0.28	99.14±0.64	99.63±0.16	99.85±0.12	**100.0 ± 0.00**	99.98±0.02	99.98±0.02	99.97 ± 0.03
6	1.77	44.30±1.92	94.33±8.50	95.28±4.79	96.90±4.09	94.02±5.20	95.33±1.92	89.09±9.86	97.24±0.90	96.58 ± 0.83	**98.44±0.49**	92.62±4.97	96.34 ± 1.24
OA (%)	90.32	95.30±0.15	92.49±5.54	96.33±1.23	95.73±1.41	97.91±0.81	97.35±0.73	97.82±0.32	98.29±0.22	98.58 ± 0.14	98.77±0.19	98.69 ± 0.32	**99.21 ± 0.17**
AA (%)	67.01	83.49±0.39	87.59±5.93	92.78±2.44	92.41±1.69	95.88±1.27	95.34±1.11	95.01±1.72	96.48±0.48	96.72 ± 0.22	97.33±0.36	96.54 ± 0.94	**97.83 ± 0.46**
$\kappa \times 100$	85.19	92.99±0.22	88.84±8.10	94.57±1.83	93.65±2.12	96.92±1.20	96.09±1.44	96.79±0.48	97.48±0.32	97.90 ± 0.20	98.19±0.28	98.06 ± 0.48	**98.75 ± 0.26**

TABLE XII

DETAILED CLASSIFICATION ACCURACY OF NILI FOSSAE ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	CSCANet	MambaHsi	HsiMamba+	DBMGNet	ours
1	97.96	97.45±0.29	96.87±2.52	98.21±1.47	97.14±2.36	96.98±1.55	97.23±1.40	98.81±0.93	98.12±0.80	99.18 ± 0.36	98.31±0.57	**99.56 ± 0.24**	99.50± 0.51
2	0.30	34.90±4.83	75.01±13.20	81.93±4.38	75.85±8.40	74.28±5.42	69.80±7.39	80.38±3.66	82.04±4.87	69.24 ± 6.04	84.67±3.39	**85.14 ± 4.81**	73.73± 5.70
3	0.00	9.66±4.68	78.51±16.68	62.35±11.41	58.31±40.09	70.99±16.34	**81.94±7.36**	64.63±12.37	60.52±22.11	48.85 ± 6.31	71.54±10.43	65.24 ± 16.33	81.55 ± 8.00
4	92.53	94.11±0.76	79.81±9.68	87.20±3.40	86.52±2.42	91.52±3.06	93.10±1.91	94.32±0.53	94.35±1.57	95.62 ± 1.01	**96.93±1.17**	95.44 ± 1.30	94.63± 1.40
5	0.00	0.22±0.41	58.22±24.70	42.65±7.99	28.45±36.08	31.68±7.16	36.33±7.80	31.81±3.53	5.58±5.64	31.86 ± 2.74	28.19±3.78	34.20 ± 2.40	**46.28± 11.43**
6	86.80	95.56±0.76	84.63±3.74	95.40±3.11	84.95±6.48	89.82±2.01	91.52±2.73	87.78±5.51	97.12±1.11	96.00 ± 2.39	95.74±1.32	97.37 ± 1.91	**98.51 ± 0.93**
7	18.82	46.77±4.97	88.40±4.57	85.14±8.04	79.56±11.74	77.05±6.04	75.74±8.68	82.40±4.90	81.79±3.46	75.63 ± 2.22	83.04±3.60	83.72 ± 1.88	**94.24 ± 3.06**
8	23.41	55.29±4.45	75.35±16.38	80.42±5.25	77.48±10.47	84.00±6.71	83.95±6.03	86.90±3.17	88.69±1.48	79.71 ± 2.93	84.74±2.84	83.86 ± 2.45	**90.53 ± 2.96**
9	93.73	93.81±0.71	91.35±3.76	95.79±1.35	91.01±2.93	95.10±0.56	94.73±1.21	94.54±0.88	95.46±0.62	96.24 ± 0.41	95.89±0.52	96.39 ± 0.42	**96.51 ± 0.53**
OA (%)	82.02	87.58±0.27	88.59±2.30	92.89±0.52	88.71±2.84	91.44±0.66	91.91±1.01	92.34±1.33	93.87±0.66	93.29 ± 0.45	94.42±0.41	95.02 ± 0.38	**95.76 ± 0.34**
AA (%)	45.92	58.64±1.05	80.91±4.84	81.01±1.79	75.47±9.87	79.05±2.23	80.48±2.20	80.17±2.52	78.19±3.12	76.92 ± 0.76	82.12±1.64	82.33 ± 1.97	**86.16 ± 2.03**
$\kappa \times 100$	76.54	84.07±0.36	85.39±2.91	91.07±0.65	86.32±3.60	89.19±0.83	88.93±1.44	90.58±1.70	92.24±0.84	91.47 ± 0.57	92.94±0.52	93.69 ± 0.48	**94.97 ± 0.44**

TABLE XIII

DETAILED CLASSIFICATION ACCURACY OF WHU-HI-HANCHUAN, ON DIFFERENT NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	SVM	RF	PyResNet	ContextualNet	SSRN	SSFTT	MATNet	SS-MSDCNet	CSCANet	MambaHsi	HsiMamba+	DBMGNet	ours
1	99.56	98.77±0.06	89.44±11.08	93.51±1.66	93.70±2.49	93.66±1.65	96.85±0.54	98.77±0.64	99.30±0.10	99.47 ± 0.16	99.51 ± 0.10	99.60±0.14	**99.86 ± 0.04**
2	89.55	91.99±0.32	93.16±2.62	91.05±3.33	91.16±5.01	93.90±2.54	94.74±2.90	95.71±1.80	96.31±0.95	97.16 ± 0.56	97.41 ± 0.20	95.53±1.12	**98.98 ± 0.28**
3	74.49	88.60±1.18	83.72±10.25	85.12±4.44	88.37±6.33	93.33±2.53	95.86±2.42	98.46±1.35	97.43±0.64	98.04 ± 2.42	98.02 ± 0.98	98.52±0.82	**99.60 ± 0.20**
4	89.87	97.55±0.29	93.67±5.57	93.57±5.48	94.36±6.98	98.25±1.65	97.92±1.10	99.53±0.51	99.35±0.25	99.11 ± 0.73	99.22 ± 0.13	99.16±0.33	**99.55 ± 0.14**
5	0.00	4.51±1.48	83.28±9.73	62.56±18.94	41.08±36.02	81.58±9.26	76.82±16.78	95.98±1.66	95.10±1.20	90.97 ± 4.88	90.48 ± 1.63	94.08±2.31	**97.84 ± 0.88**
6	0.80	9.84±1.86	67.41±12.86	61.92±9.39	64.29±11.57	78.47±6.72	80.11±9.08	82.75±7.00	80.38±1.59	85.21 ± 8.10	87.53 ± 1.64	86.43±1.33	**92.93 ± 1.16**
7	90.43	89.83±0.81	74.65±10.95	81.82±9.75	82.02±9.24	84.52±7.85	88.62±9.05	93.33±2.65	**94.43±1.20**	90.69 ± 6.54	92.27 ± 0.59	93.91±0.53	93.98 ± 0.53
8	90.77	86.88±0.60	88.21±7.96	84.16±3.96	85.99±4.31	90.48±2.99	93.27±3.75	94.63±2.75	95.46±0.99	97.01 ± 0.94	97.69 ± 0.40	97.04±0.33	**97.83 ± 0.29**
9	35.56	72.87±1.50	78.87±6.74	77.68±8.08	88.20±5.57	84.96±6.83	88.75±7.40	96.02±1.72	96.80±0.55	98.22 ± 0.43	**98.64 ± 0.34**	96.63±6.21	**98.64 ± 0.25**
10	77.06	89.90±1.19	92.76±12.83	96.69±1.90	98.23±1.21	98.04±0.98	98.62±0.87	98.40±0.60	98.55±0.29	98.63 ± 1.61	98.86 ± 0.32	**99.47±0.22**	98.90 ± 0.20
11	89.60	98.05±0.18	85.13±11.11	92.23±4.24	93.25±2.42	94.19±2.14	93.99±3.04	95.62±4.36	99.09±0.23	98.55 ± 0.41	98.38 ± 0.20	98.50±0.24	**99.25 ± 0.18**
12	0.03	29.66±1.76	75.41±6.90	59.89±8.07	43.83±23.68	81.87±5.11	85.14±5.75	85.95±9.71	94.51±1.40	92.00 ± 11.25	93.40 ± 1.71	93.72±3.68	**99.07 ± 0.58**
13	46.12	56.51±0.93	77.38±5.30	72.15±12.51	74.63±11.45	84.06±4.45	86.61±4.83	78.11±13.64	84.99±1.34	87.94 ± 2.54	87.98 ± 1.05	87.70±1.18	**91.71 ± 0.84**
14	85.06	91.19±0.28	84.30±6.62	87.17±1.93	87.44±5.96	93.55±2.73	95.55±3.50	97.29±1.20	96.49±0.36	97.66 ± 0.87	97.78 ± 0.27	97.66±0.22	**98.22 ± 0.36**
15	0.70	0.78±0.44	78.76±17.11	79.37±11.56	67.95±39.47	88.63±14.55	78.84±17.47	64.36±6.12	84.53±3.37	90.01 ± 5.55	**90.49 ± 3.32**	88.22±2.68	82.33 ± 2.41
16	99.78	98.93±0.11	98.51±1.13	99.34±0.50	99.86±0.10	99.77±0.07	99.51±0.57	99.88±0.07	99.70±0.06	99.87 ± 0.14	99.86 ± 0.03	99.88±0.04	**99.95 ± 0.04**
OA (%)	85.79	89.92±0.09	89.86±3.74	90.75±1.02	91.51±1.38	94.15±0.57	95.48±1.34	96.63±1.24	97.37±0.15	97.83 ± 0.99	98.02 ± 0.11	97.82 ± 0.41	**98.76 ± 0.04**
AA (%)	60.59	69.12±0.19	84.04±2.88	82.39±2.10	80.90±5.70	89.95±1.20	90.70±3.34	92.18±2.14	94.53±0.31	95.03 ± 2.80	95.47 ± 0.41	95.38 ± 0.65	**96.79 ± 0.25**
$\kappa \times 100$	83.20	88.15±0.11	86.72±4.47	89.09±1.19	90.13±1.63	93.00±0.66	94.57±1.57	96.05±1.46	96.92±0.18	97.45 ± 1.16	97.69 ± 0.13	97.45 ± 0.47	**98.57 ± 0.09**

influence of redundant responses and enhance discriminative representations.

Classification maps are presented in Figs. 7–9. On the Holden Crater and Nili Fossae datasets, GCEMNet produces more spatially consistent mineral maps and reduces local misclassification noise. On the WHU-Hi-HanChuan dataset, it better preserves crop-field boundaries and suppresses confusion among similar crop and background categories, leading to clearer regional structures. The visual results are consistent with the quantitative comparisons, further demonstrating the spatial consistency and robustness of the proposed method in challenging classification scenarios.

F. Ablation Analysis

To verify the effectiveness of each key component, we perform unified ablation studies on all three datasets. For the GC module, we remove the grouping structure and directly feed global features into subsequent layers as the baseline. For the EM module, we replace it with 1-D convolution and traditional attention mechanisms for comparison. For the CSAF module, we conduct control experiments by removing this component entirely. For the CFST module, we design comparison schemes including direct feature concatenation and nonseparation structures.

TABLE XIV

ABLATION RESULTS OF DIFFERENT MODULES ON THREE DATASETS

Dataset	GC	EM	CSAF	CFST	OA (%)	AA (%)	$\kappa \times 100$
Holden	×	√	√	√	98.11 ± 0.39	95.28 ± 1.08	97.21 ± 0.58
	√	×	√	√	98.01 ± 0.67	94.82 ± 1.26	97.06 ± 0.99
	√	√	×	√	98.02 ± 0.39	94.79 ± 1.24	97.08 ± 0.57
	√	√	√	×	97.65 ± 0.35	93.43 ± 1.04	96.54 ± 0.51
	√	√	×	×	98.34 ± 0.22	95.79 ± 0.85	97.55 ± 0.33
	×	×	√	√	97.42 ± 0.37	91.86 ± 1.55	96.19 ± 0.55
	√	√	√	√	**99.21 ± 0.17**	**97.83 ± 0.46**	**98.75 ± 0.26**
Nili	×	√	√	√	95.48 ± 0.62	84.96 ± 3.19	94.28 ± 0.79
	√	×	√	√	94.97 ± 0.33	77.36 ± 1.95	92.98 ± 0.47
	√	√	×	√	95.66 ± 0.50	85.09 ± 1.80	94.50 ± 0.63
	√	√	√	×	93.79 ± 0.41	79.13 ± 1.97	92.13 ± 0.53
	√	√	×	×	95.40 ± 0.74	85.06 ± 2.18	94.17 ± 0.94
	×	×	√	√	94.48 ± 0.59	80.19 ± 2.63	93.00 ± 0.75
	√	√	√	√	**95.76 ± 0.34**	**86.16 ± 2.03**	**94.97 ± 0.44**
Hanchuan	×	√	√	√	98.66 ± 0.02	96.51 ± 0.04	98.38 ± 0.05
	√	×	√	√	98.60 ± 0.06	96.20 ± 0.25	98.31 ± 0.14
	√	√	×	√	98.19 ± 0.11	95.73 ± 0.30	97.83 ± 0.24
	√	√	√	×	94.23 ± 0.14	78.03 ± 0.84	93.17 ± 0.33
	√	√	×	×	98.58 ± 0.04	96.34 ± 0.31	98.29 ± 0.10
	×	×	√	√	95.43 ± 0.15	87.27 ± 0.83	95.19 ± 0.33
	√	√	√	√	**98.76 ± 0.04**	**96.79 ± 0.25**	**98.57 ± 0.09**

Quantitative results show that the complete GCEMNet achieves the best overall performance on all three datasets. Removing each core component causes different degrees of

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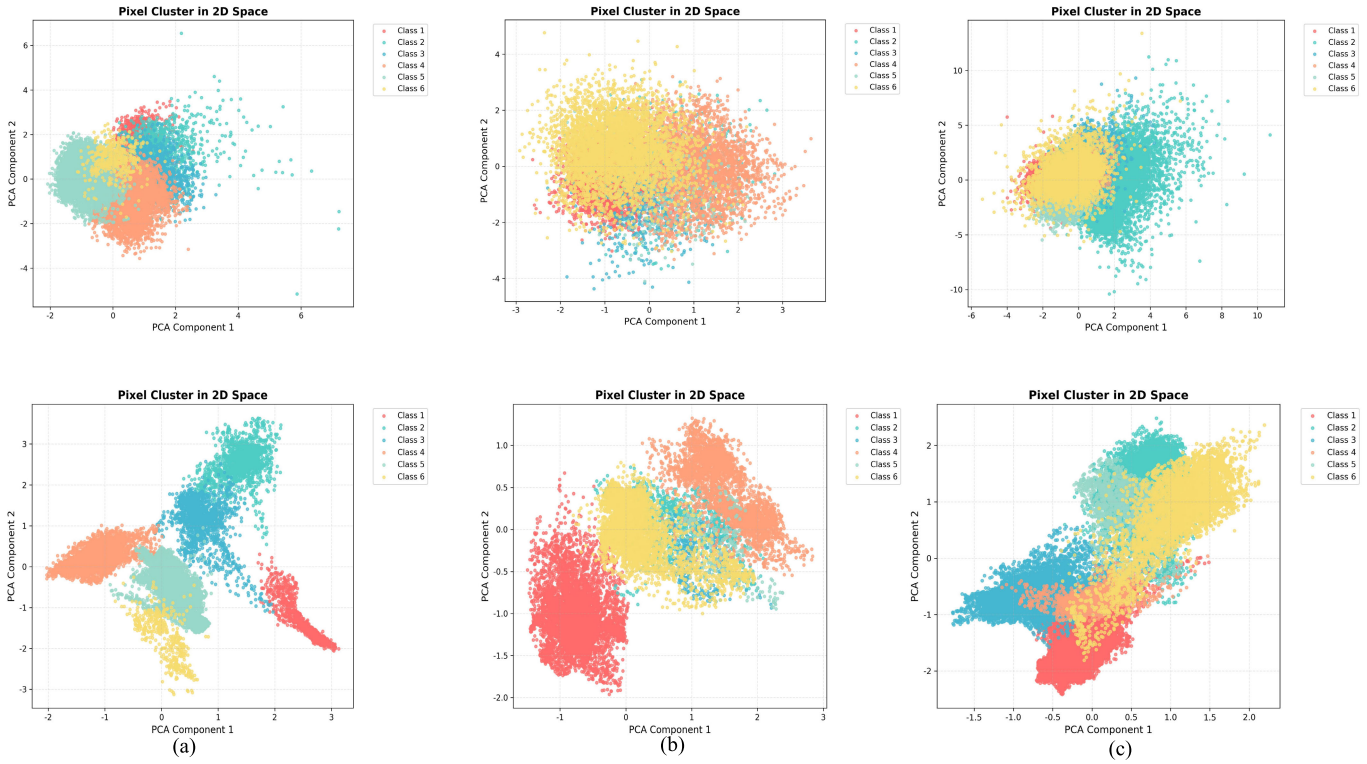


Fig. 6. Visualization of feature distributions after PCA and k-means clustering. (a) Holden Crater. (b) Nili Fossae. (c) WHU-Hi-HanChuan.

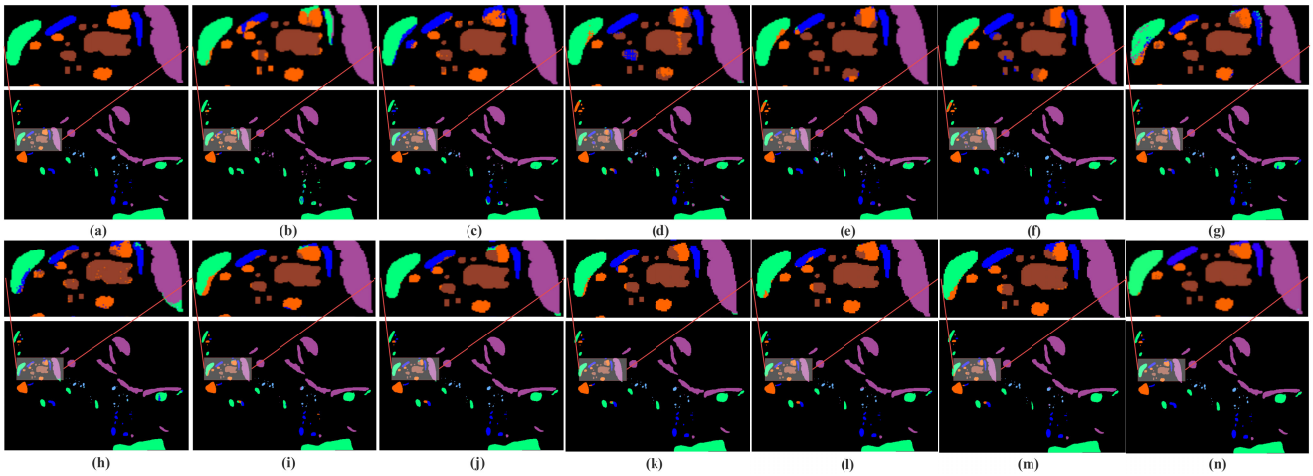


Fig. 7. Classification maps of different methods on the Holden Crater dataset. (a) Ground truth. (b) SVM [13]. (c) RF [14]. (d) PyResNet [43]. (e) ContextualNet [44]. (f) SSRN [45]. (g) SSFTT [46]. (h) MATNet [47]. (i) SS-MSDCNet [48]. (j) CSCANet [49]. (k) MambaHsi [50]. (l) HsiMamba+ [51]. (m) DBMGNet [52]. (n) GCEMNet (ours).

performance degradation, indicating that GC, EM, CSAF, and CFST provide complementary contributions to the final classification capability under different spectral and spatial conditions. Specifically, removing GC reduces OA by 1.10%, 0.28%, and 0.10% on Holden Crater, Nili Fossae, and WHU-Hi-HanChuan, respectively, demonstrating that grouped cascaded propagation helps reduce redundant channel interference and improve subband feature modeling. Replacing EM weakens the modeling of long-range spectral–spatial dependencies, especially in terms of AA, which indicates its importance for fine-grained recognition of spectrally similar and minority categories. Without CSAF, the model shows reduced discriminability in shallow feature extraction, confirming that collaborative channel–spatial attention provides useful low-level representations. Removing CFST from the complete configuration produces the largest marginal degradation, especially on the WHU-Hi-HanChuan dataset, where OA decreases by 4.53%, verifying the importance of matrix-guided feature separation and cross-feature interaction for redundancy suppression and complementary information enhancement. These

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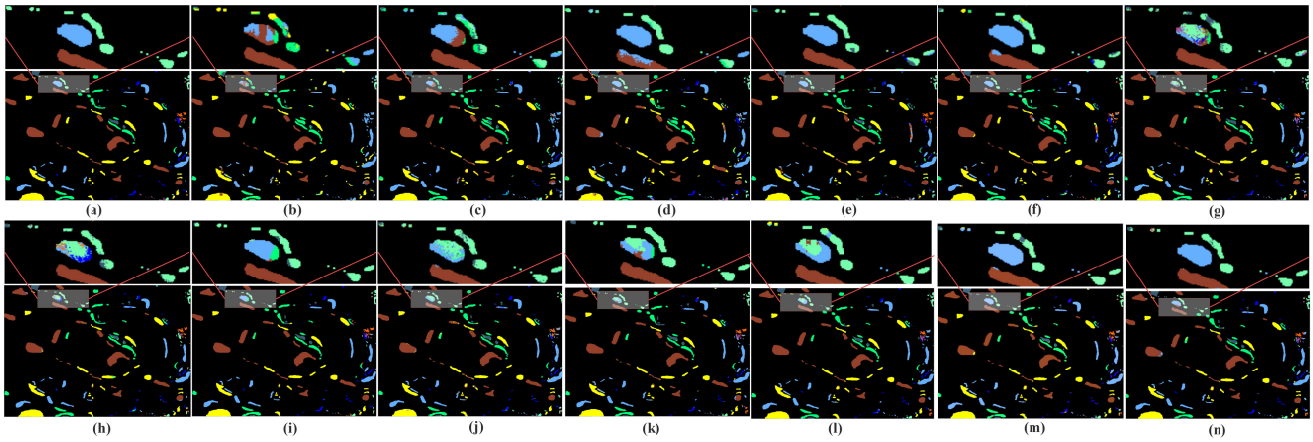


Fig. 8. Classification maps of different methods on the Nili Fossae dataset. (a) Ground truth. (b) SVM [13]. (c) RF [14]. (d) PyResNet [43]. (e) ContextualNet [44]. (f) SSRN [45]. (g) SSFTT [46]. (h) MATNet [47]. (i) SS-MSDCNet [48]. (j) CSCANet [49]. (k) MambaHsi [50]. (l) HsiMamba+ [51]. (m) DBMGNet [52]. (n) GCEMNet (ours).

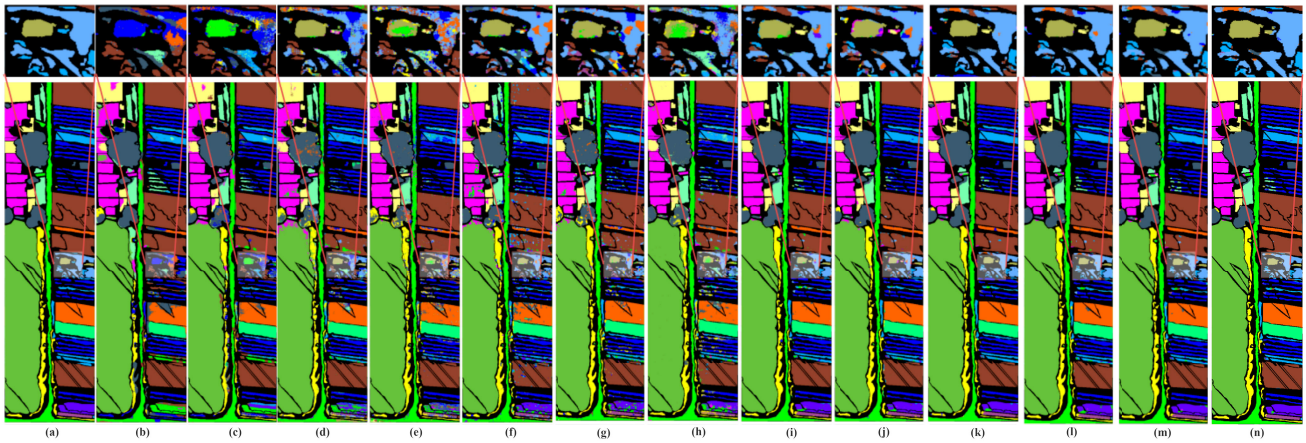


Fig. 9. Classification maps of different methods on the WHU-Hi-HanChuan dataset. (a) Ground truth. (b) SVM [13]. (c) RF [14]. (d) PyResNet [43]. (e) ContextualNet [44]. (f) SSRN [45]. (g) SSFTT [46]. (h) MATNet [47]. (i) SS-MSDCNet [48]. (j) CSCANet [49]. (k) MambaHsi [50]. (l) HsiMamba+ [51]. (m) DBMGNet [52]. (n) GCEMNet (ours).

results confirm the synergistic contribution of the proposed modules rather than the dominance of a single component. Detailed statistical information is presented in Table XIV.

G. Efficiency Analysis

To evaluate the practical applicability of GCEMNet, we compare its classification performance and computational complexity with representative HSI classification methods on the Nili Fossae dataset. Model size and computational complexity are measured by parameter count and floating-point operations (FLOPs), respectively. Fig. 10 illustrates the relationship between classification accuracy and computational complexity, where the horizontal axis denotes FLOPs, the vertical axis denotes OA, and the bubble size and color jointly indicate the parameter count.

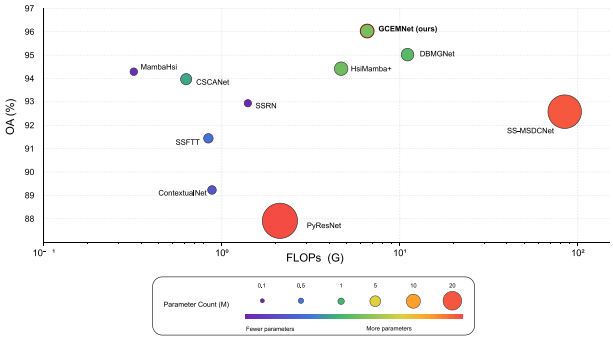


Fig. 10. Classification accuracy and computational complexity of different methods on the Nili Fossae dataset. The horizontal axis denotes FLOPs, the vertical axis denotes OA, and the bubble size and color indicate the parameter count.

As shown in Fig. 10, lightweight models such as MambaHsi, SSRN, and SSFTT have relatively low computational cost or small parameter size, but their classification accuracy is lower than that of GCEMNet. Some larger models, such as PyResNet and SS-MSDCNet, involve a larger number of parameters or higher FLOPs, yet do not achieve competitive OA. In contrast, GCEMNet achieves the highest OA with moderate FLOPs

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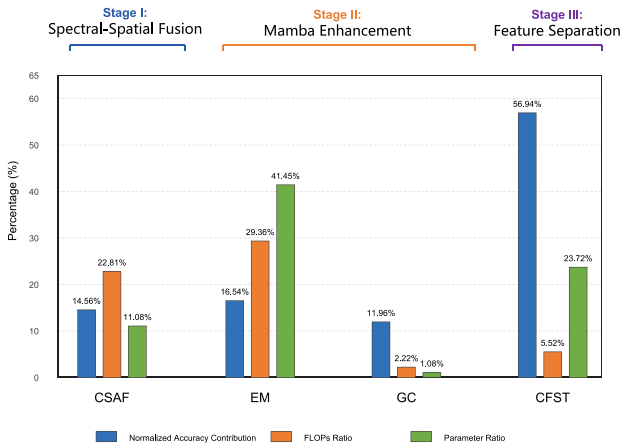


Fig. 11. Module-level contribution and computational complexity analysis of GCEMNet. The normalized accuracy contribution, FLOPs ratio, and parameter ratio are compared among CSAF, EM, GC, and CFST.

and a controlled parameter scale, demonstrating a favorable balance between classification accuracy and computational complexity under severe spectral confusion.

Fig. 11 further analyzes the module-level contribution and computational complexity of GCEMNet. The results show that EM and CSAF account for the main computational cost, since they are responsible for Mamba-based enhancement and shallow feature fusion, respectively. GC introduces only a small amount of additional computation and parameters, while CFST contributes the largest normalized accuracy improvement with a relatively low FLOPs ratio. These results suggest that the performance gain of GCEMNet mainly comes from effective module collaboration rather than simply increasing model complexity.

V. Conclusion

In this article, we proposed GCEMNet for HSI classification in spectrally similar scenarios. The network follows a progressive feature pipeline composed of feature fusion extraction, grouped cascaded Mamba enhancement, and matrix-guided separation. In the first stage, CSAF enhances shallow spectral and spatial representations through channel and spatial attention, while EM models channel-wise and spatial sequential dependencies to capture discriminative cues. In the second stage, GCMF, built upon EM units, performs grouped enhancement and cascaded aggregation between local and global context. The spectral-guided Mamba refinement module injects spectral guidance into spatial sequence modeling, improving sensitivity to texture variations among similar categories.

In the feature separation stage, CFST is introduced to reduce redundant spectral and spatial responses and enhance complementary information. Specifically, MFS constructs spectral and spatial adjacency matrices to obtain correlation-aware separated representations, while CFT performs bidirectional cross-feature interaction between the separated features. Through the collaboration of CSAF, EM, GCMF, spectral-guided refinement, and CFST, GCEMNet improves shallow fusion, dependency modeling, local and global feature aggregation, and redundancy suppression within a unified framework. This design is useful for categories with overlapping spectra, similar spatial structures, and limited training samples.

Experiments on two mineral and one agricultural HSI datasets demonstrate that GCEMNet achieves superior classification performance over traditional, CNN-based, Transformer-based, and Mamba-based methods. Parameter sensitivity analysis, module replacement experiments, ablation studies, visualization results, and complexity analysis verify the effectiveness of the proposed modules for spectrally similar scenarios and scenarios with class imbalance. Results show that GCEMNet achieves a favorable balance between accuracy and complexity. Nevertheless, minority or highly confused categories remain challenging under sample imbalance. Future work will focus on lightweight optimization, more robust spectral representation learning, and efficient deployment for real-time and small-sample HSI classification.

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